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Hyung

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(54) **METHOD FOR FORMING
SEMICONDUCTOR DEVICE HAVING AN
AIR GAP BETWEEN A CONTACT PAD AND
A SIDEWALL OF CONTACT HOLE**

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8, 2021, now Pat. No. 12,057,396.

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Nov. 25, 2020 (CN) 202022770473.3

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H01L 23/528 (2006.01)
H10D 62/10 (2025.01)

(52) **U.S. Cl.**
CPC **H01L 23/5283** (2013.01); **H01L 21/76895**
(2013.01); **H10D 62/115** (2025.01)

(58) **Field of Classification Search**
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21/76897; H01L 23/5283; H10D 62/115;
H10B 12/0335; H10B 12/315; H10B
12/482
USPC 438/262
See application file for complete search history.

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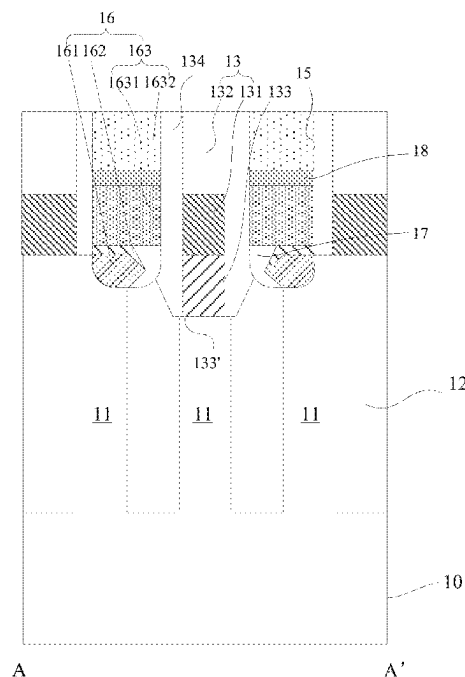
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Primary Examiner — Hoai V Pham

(57) **ABSTRACT**

Disclosed are a semiconductor device and a method for preparing a semiconductor device. The semiconductor device is provided with contact pad structures in contact holes. Each of the contact pad structures is configured to comprise a first contact pad, a second contact pad adaptively covering the first contact pad, and a contact plug located on the second contact pad. The first contact pad is in full contact with an active region in a substrate. In addition, an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole.

10 Claims, 21 Drawing Sheets



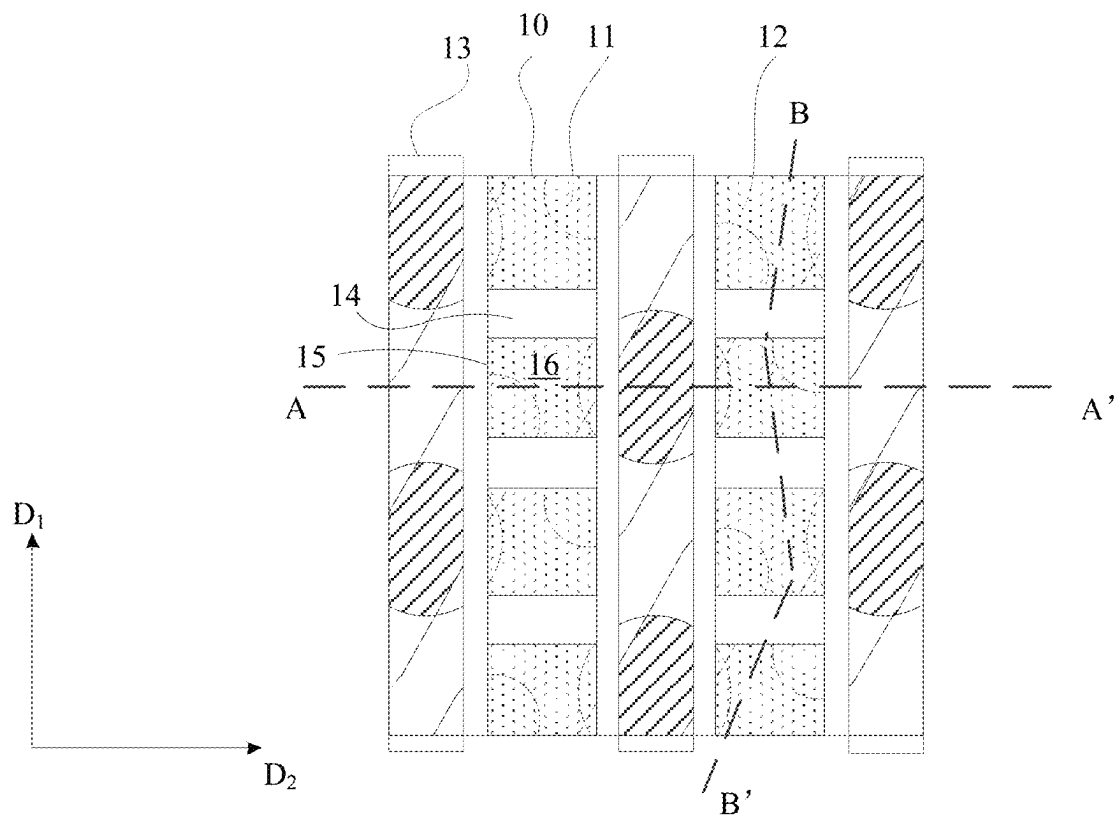


Figure 1

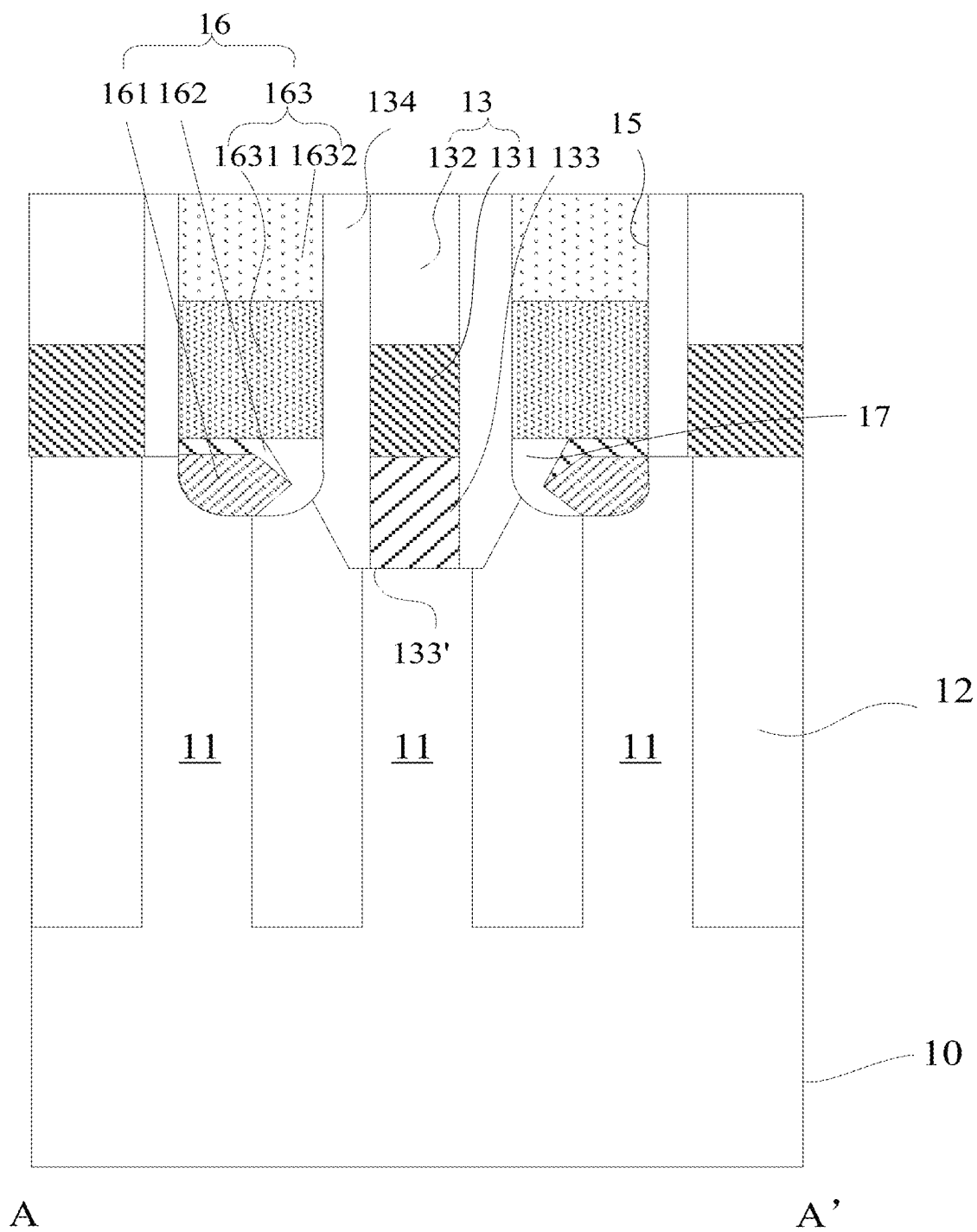


Figure 2A

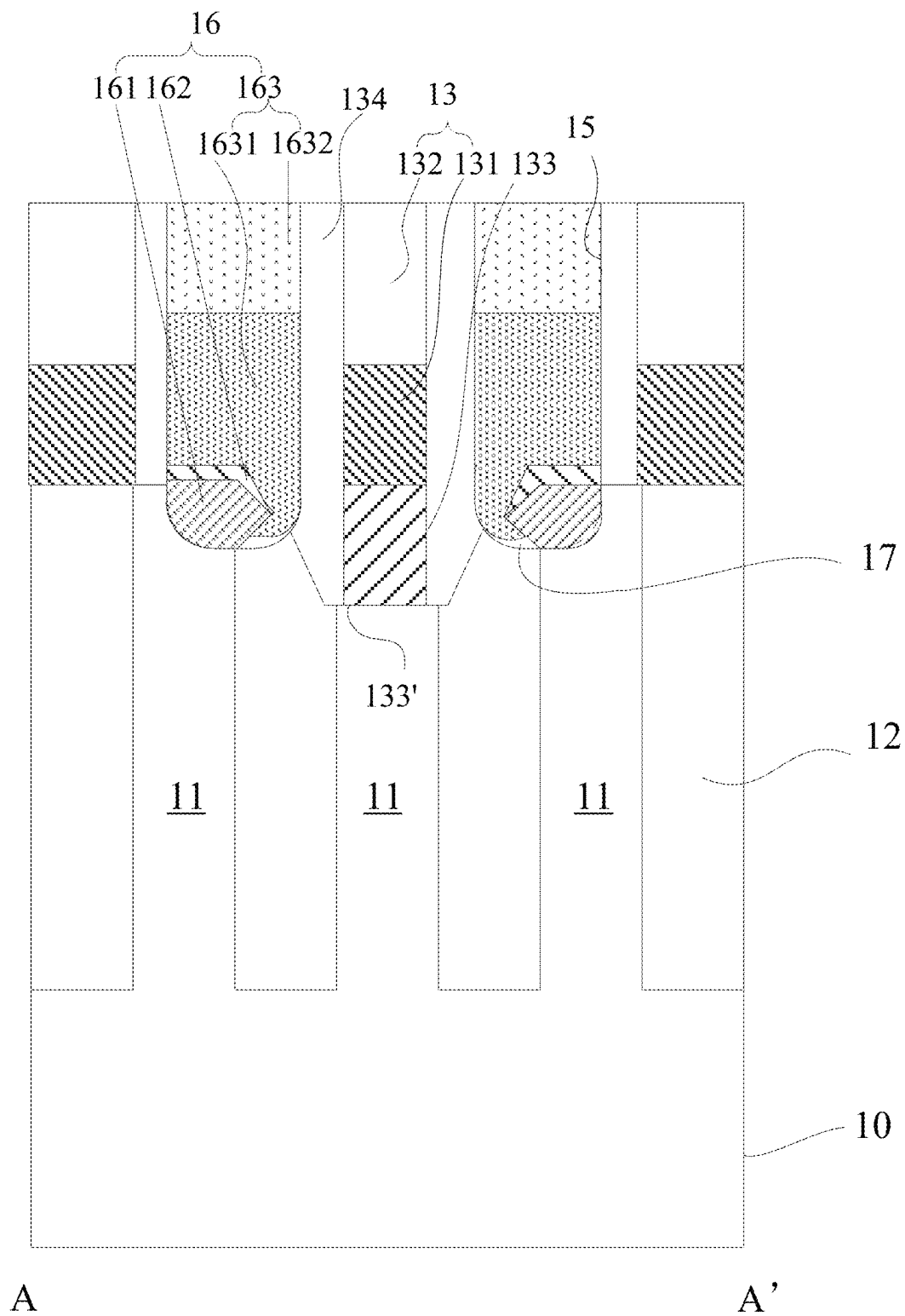


Figure 2B

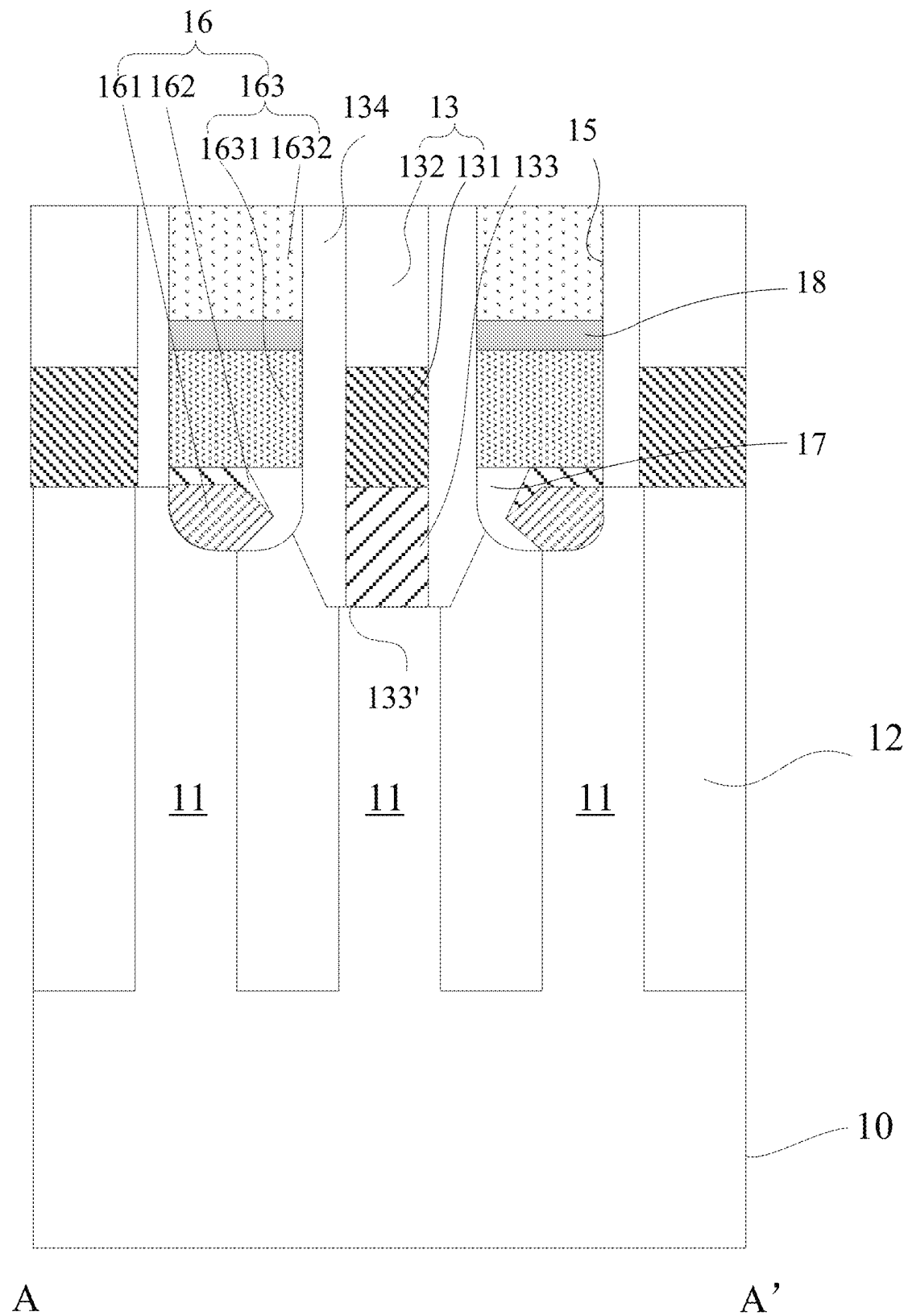


Figure 2C

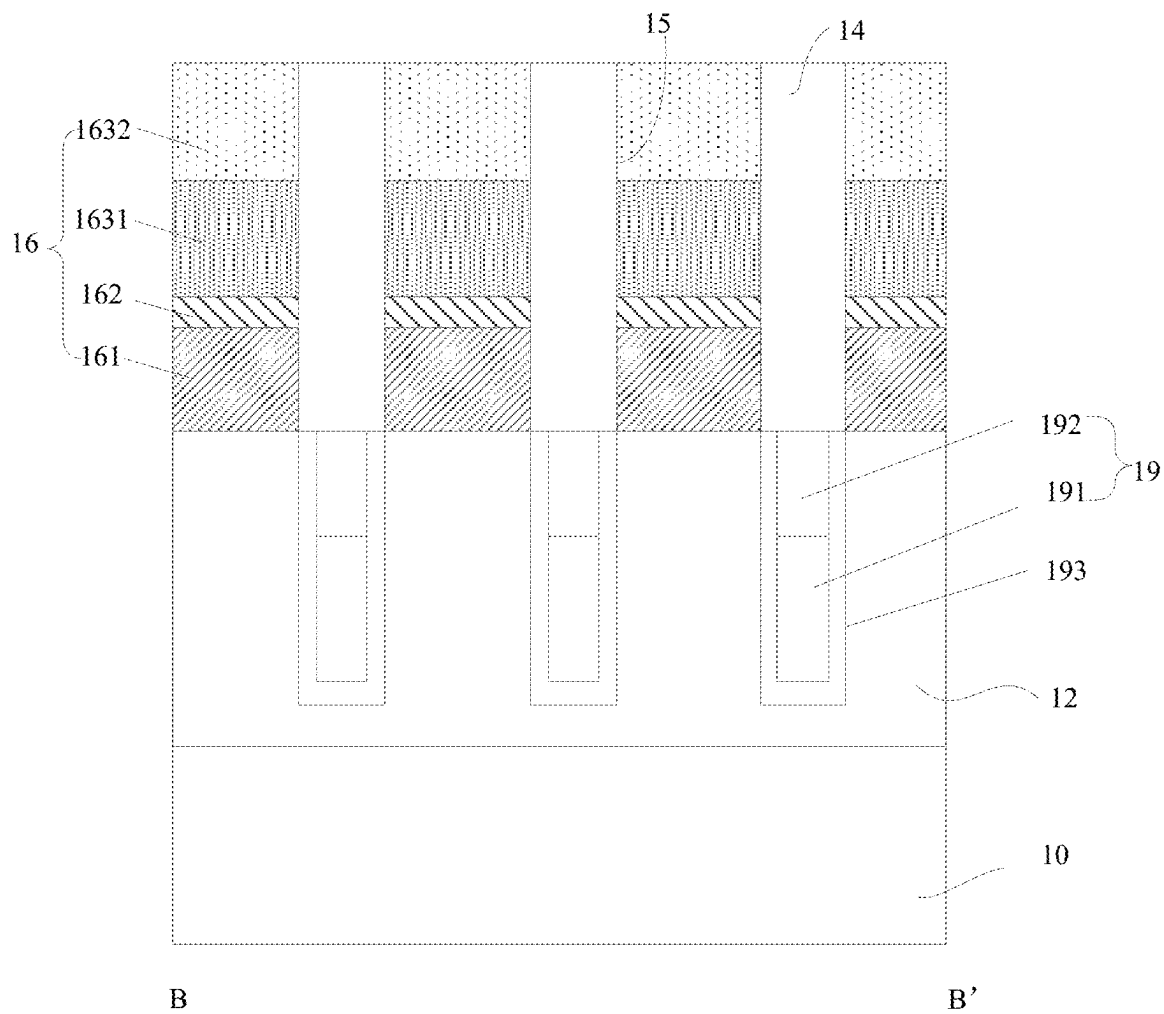


Figure 3

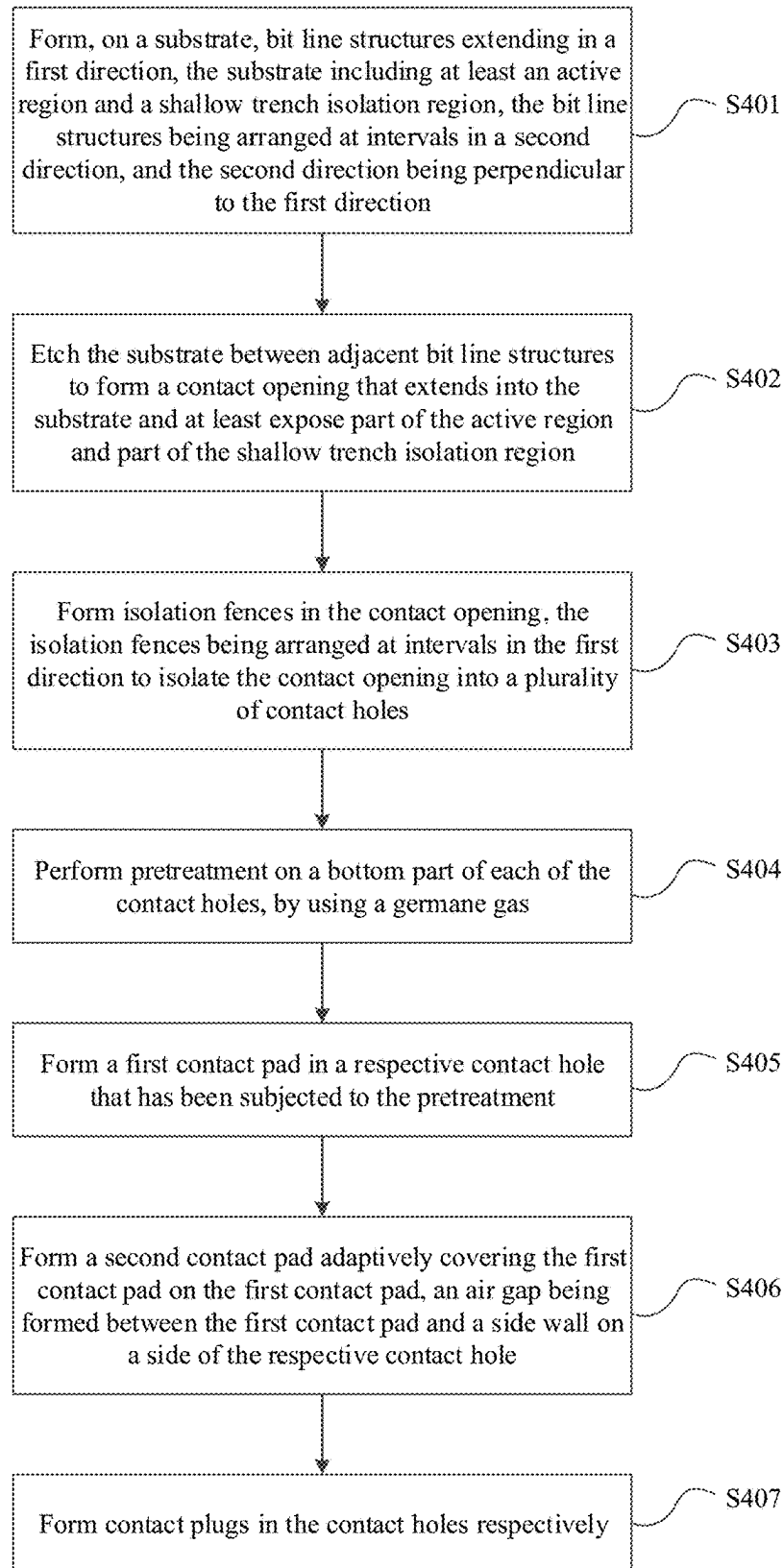


Figure 4

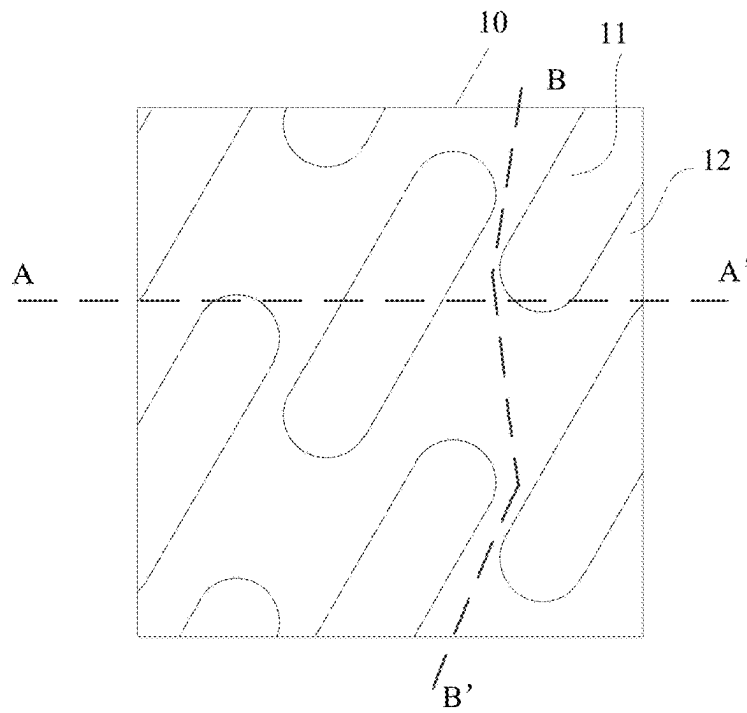


Figure 5A

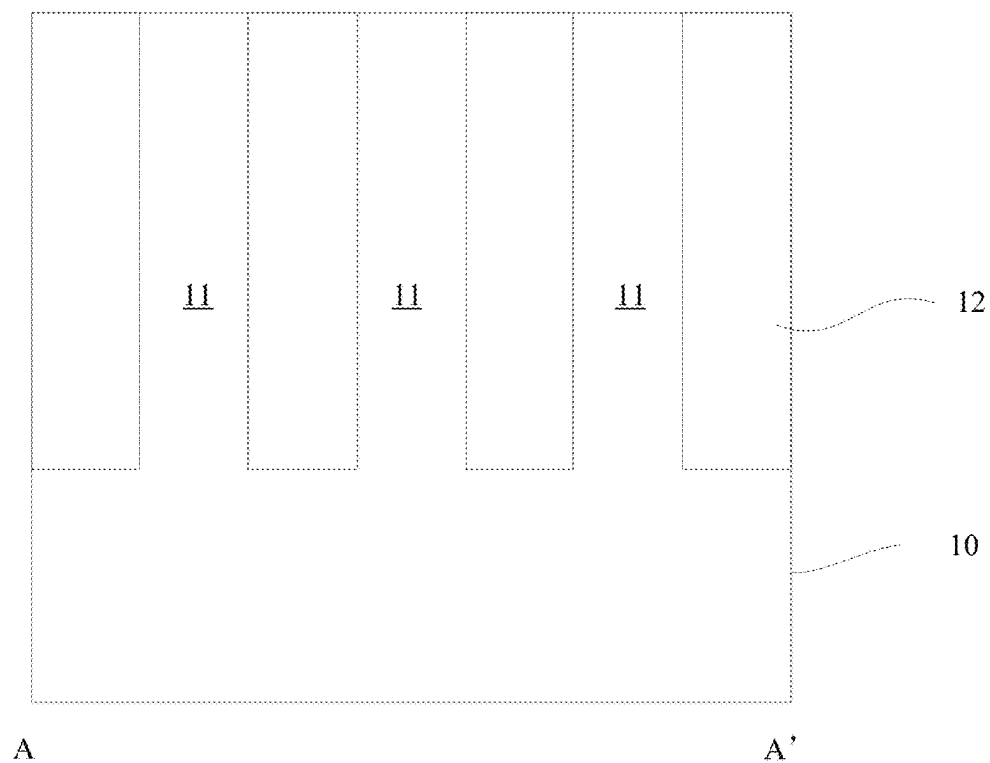


Figure 5B

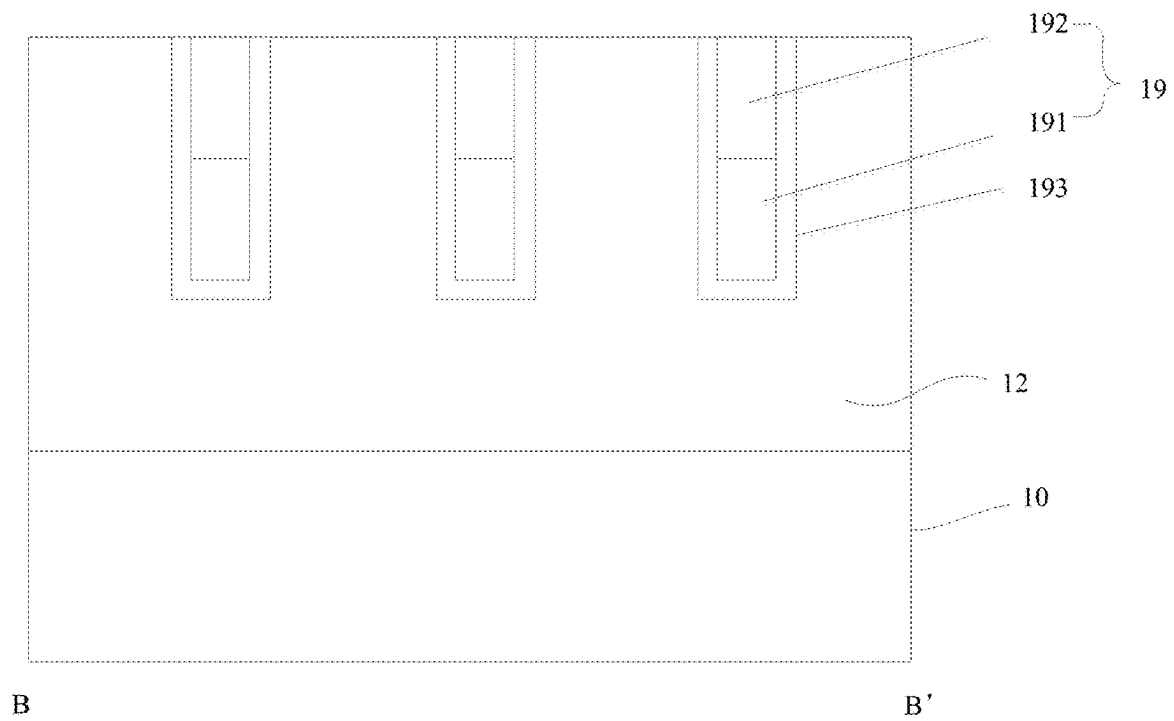


Figure 5C

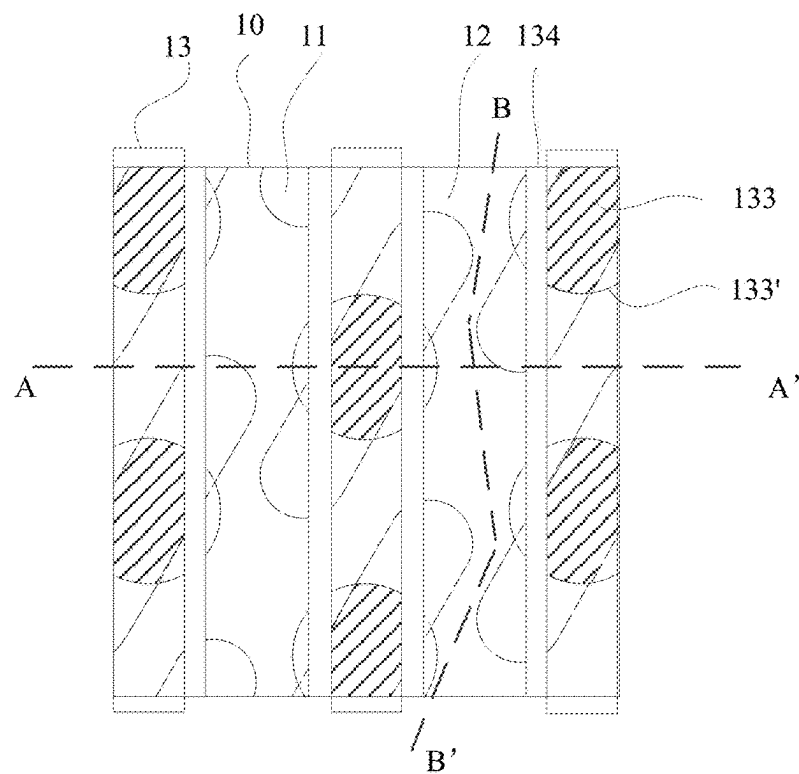


Figure 6A

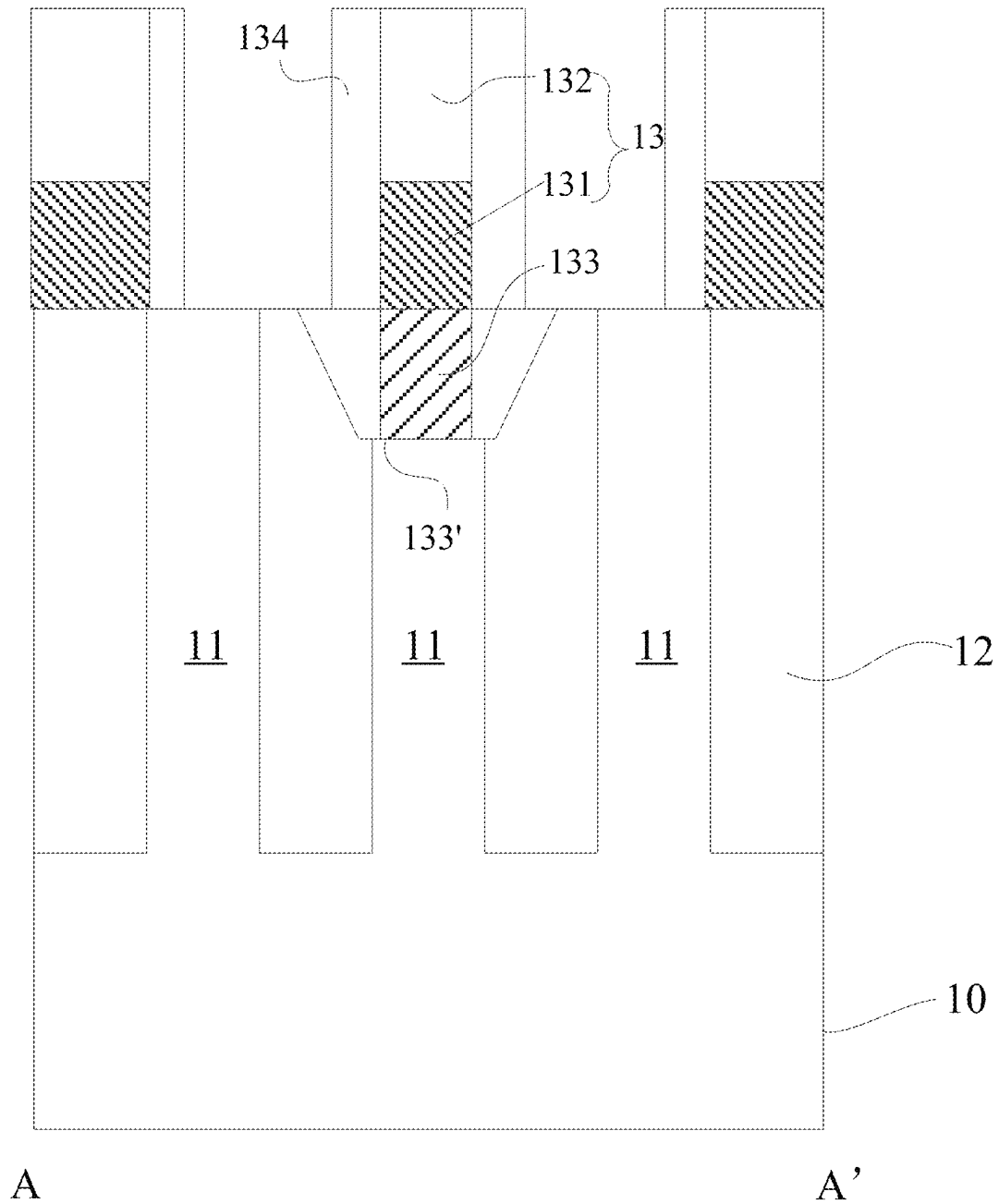


Figure 6B

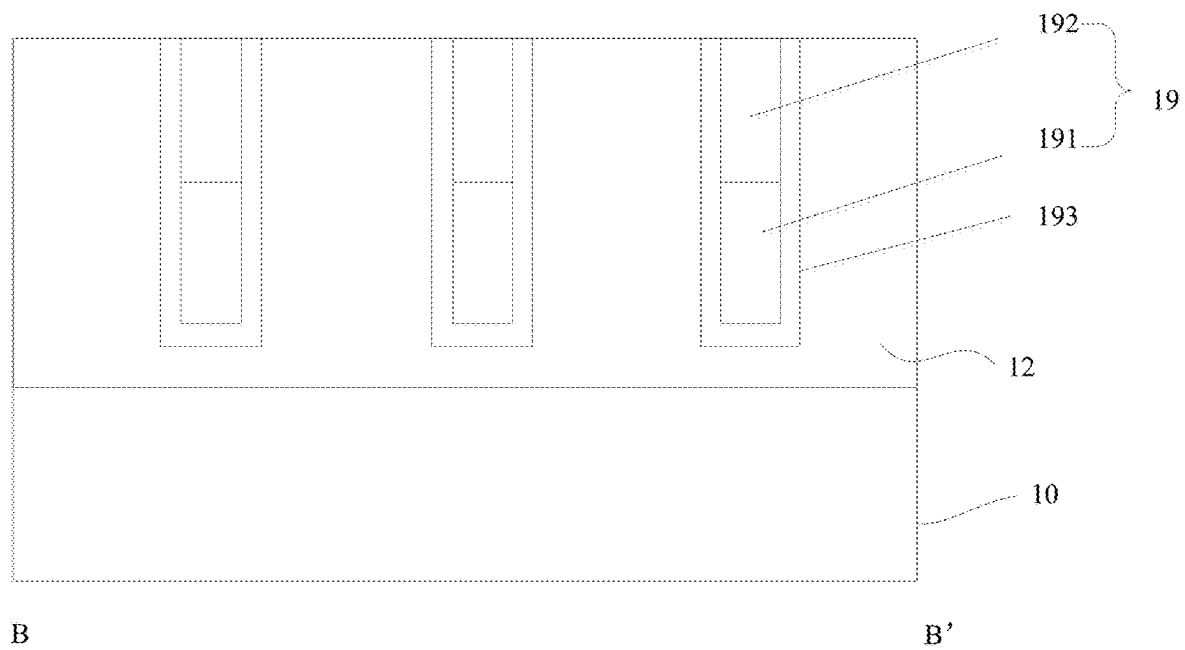


Figure 6C

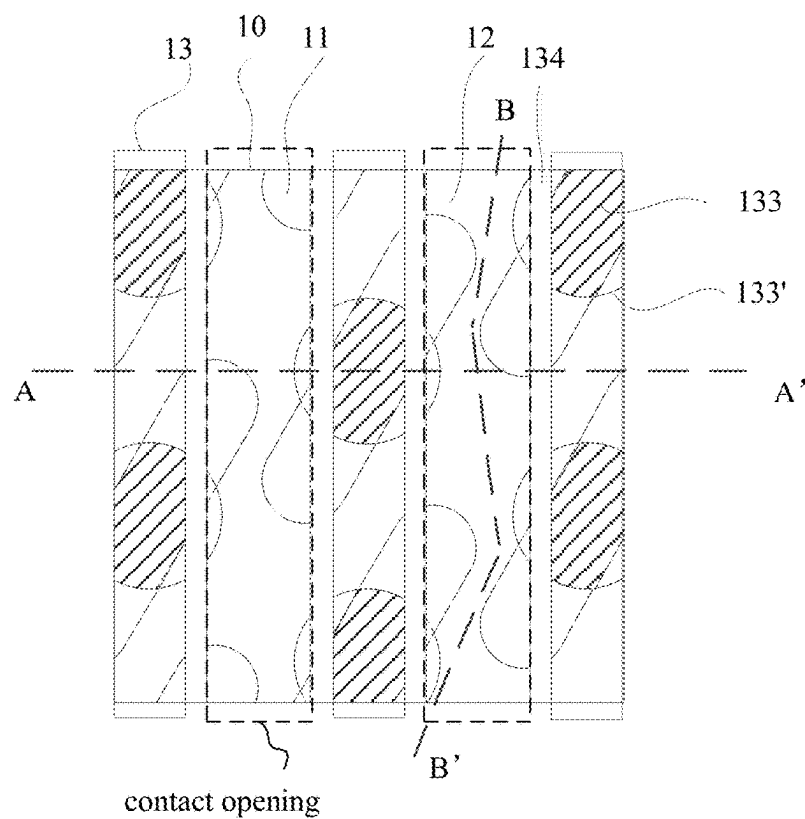


Figure 7A

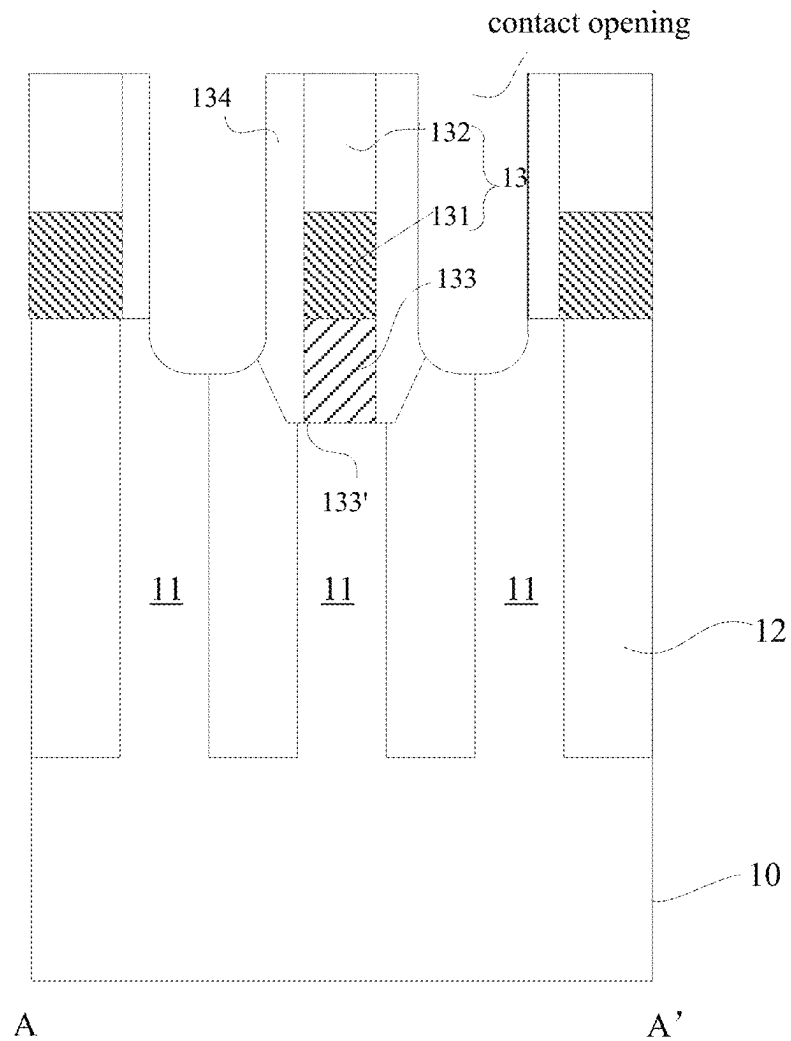


Figure 7B

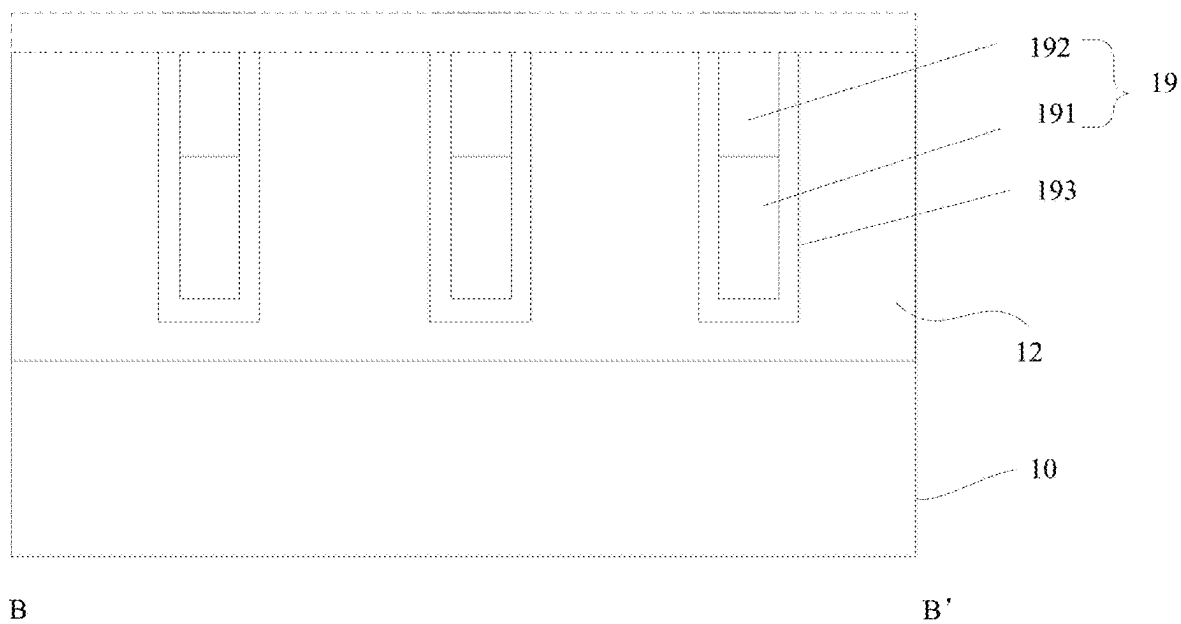


Figure 7C

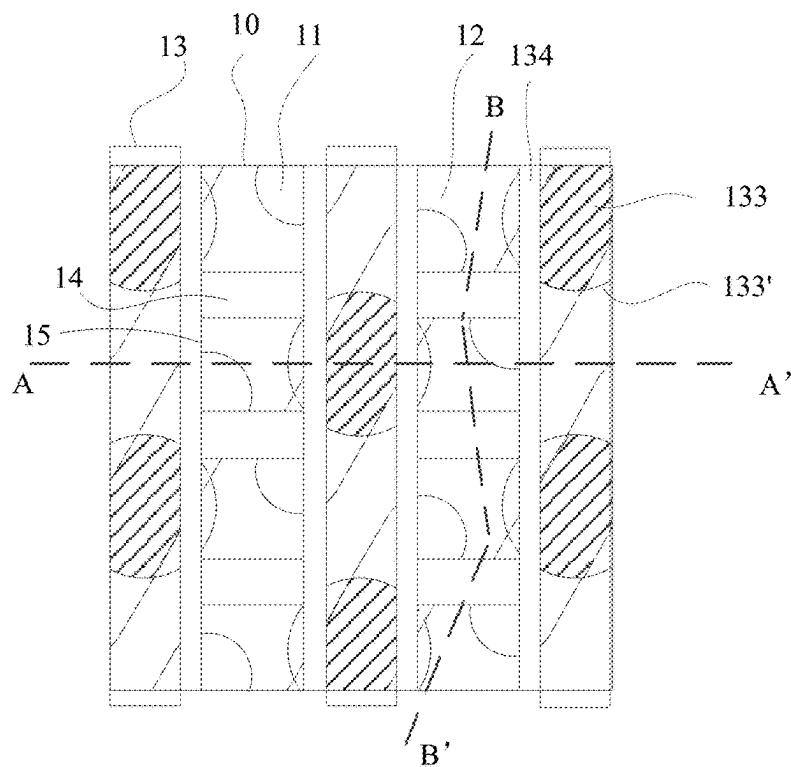


Figure 8A

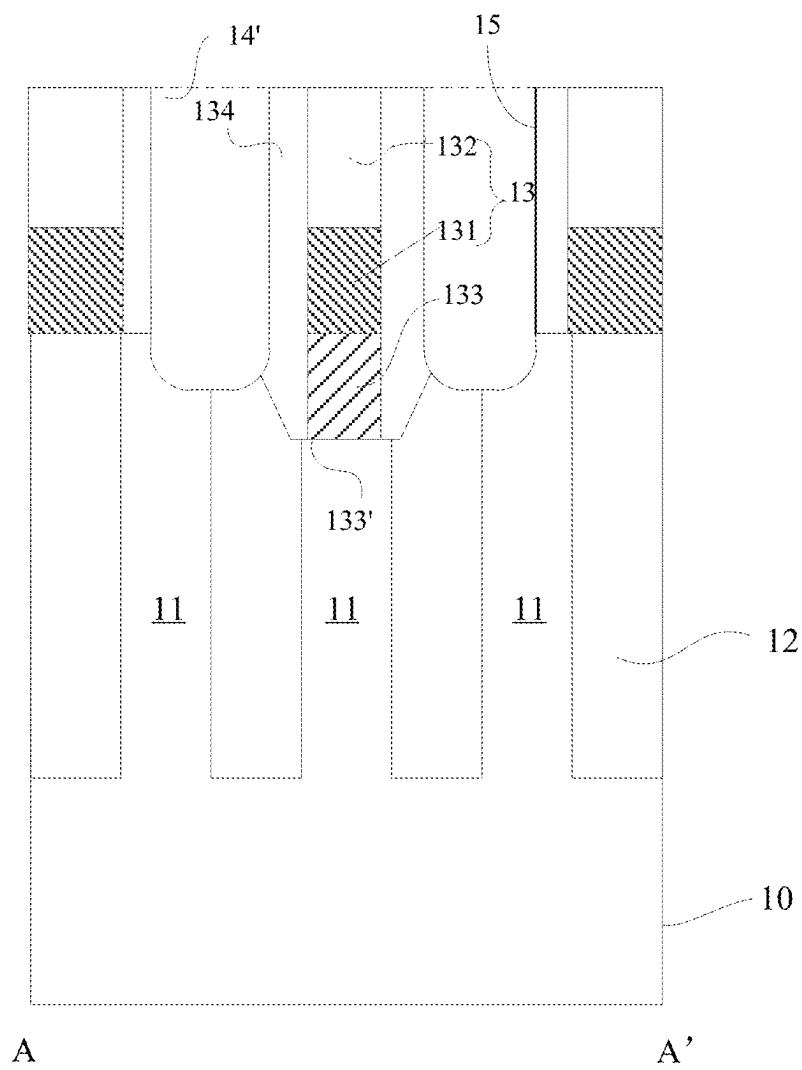


Figure 8B

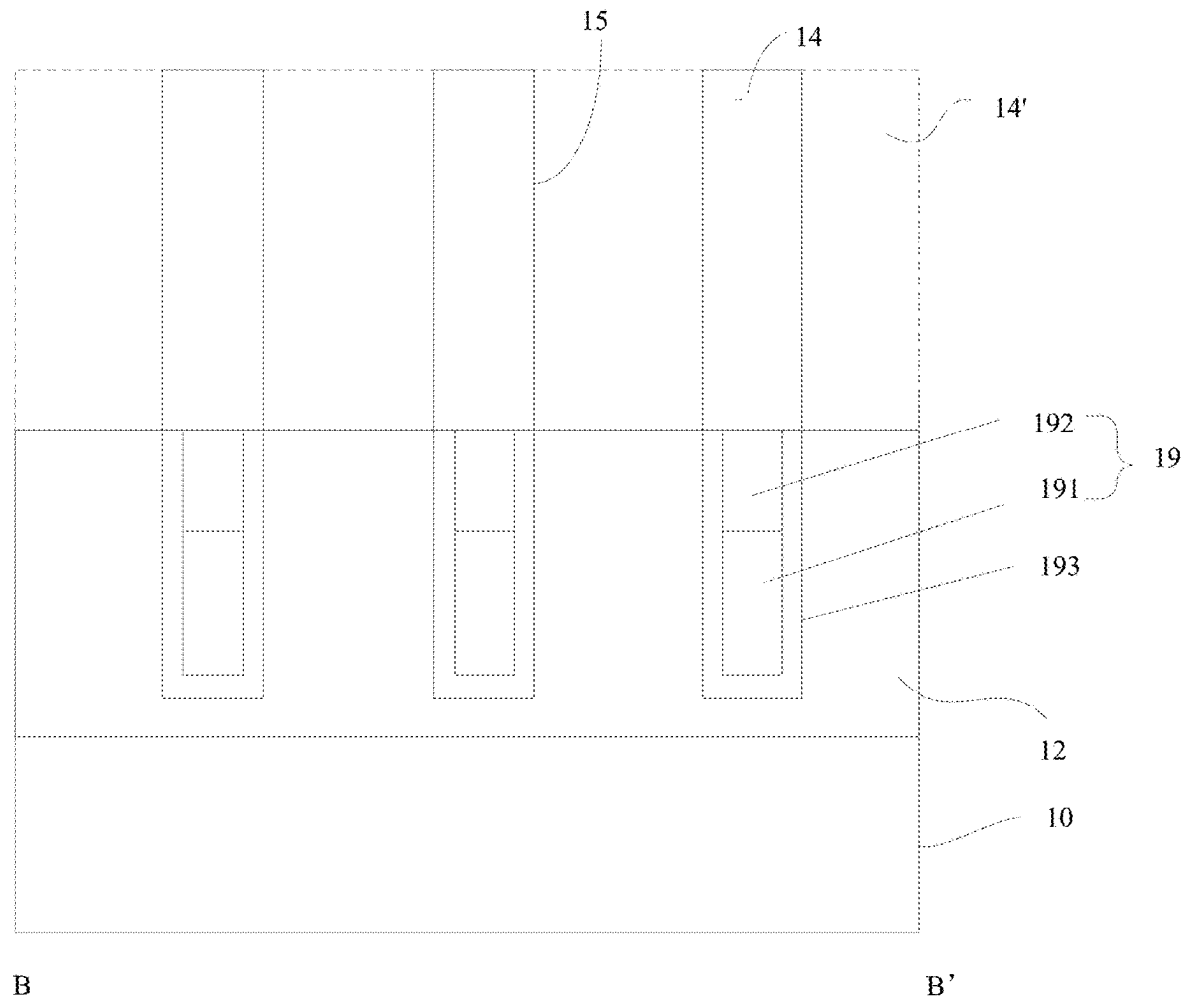


Figure 8C

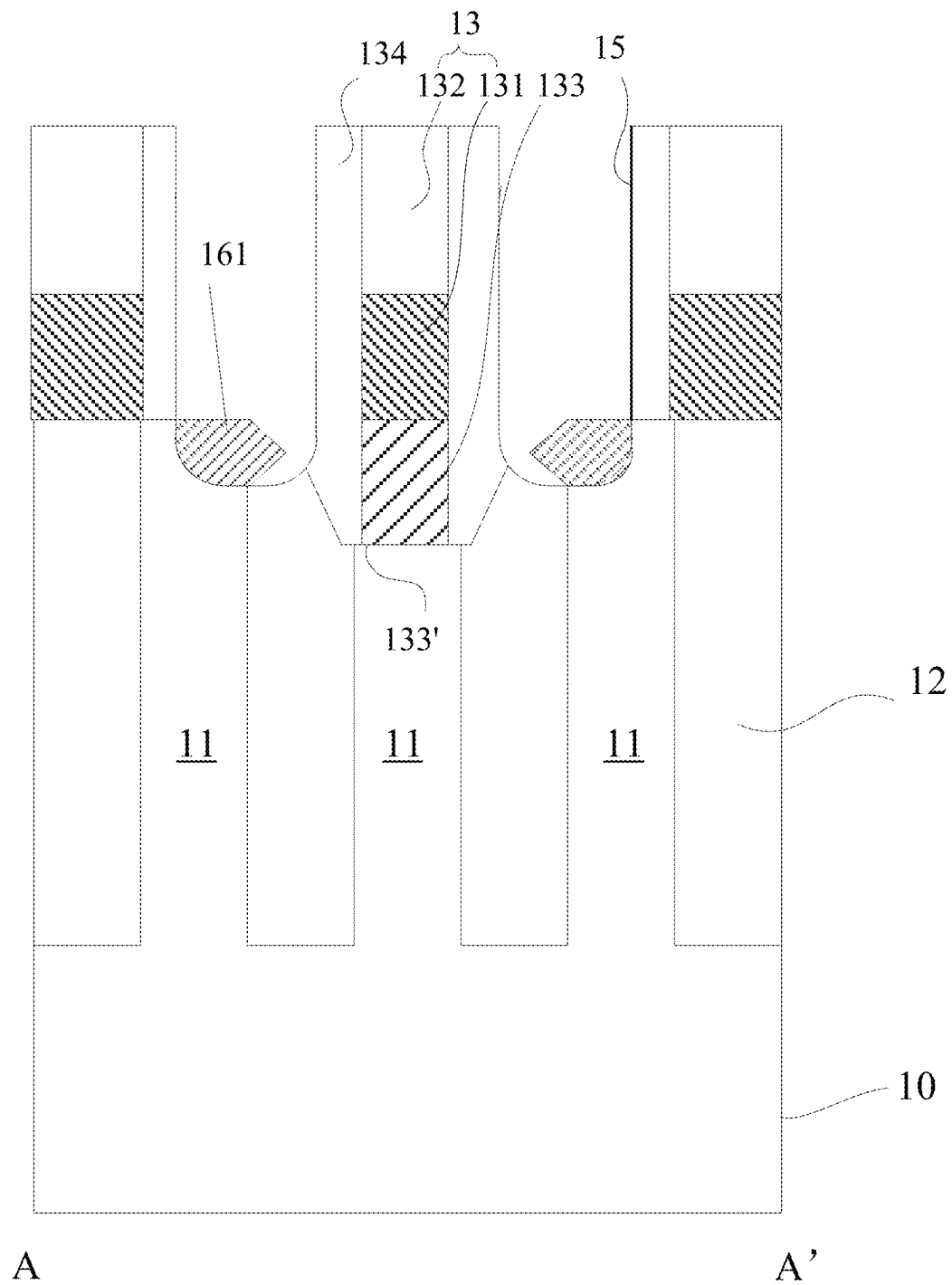


Figure 9A

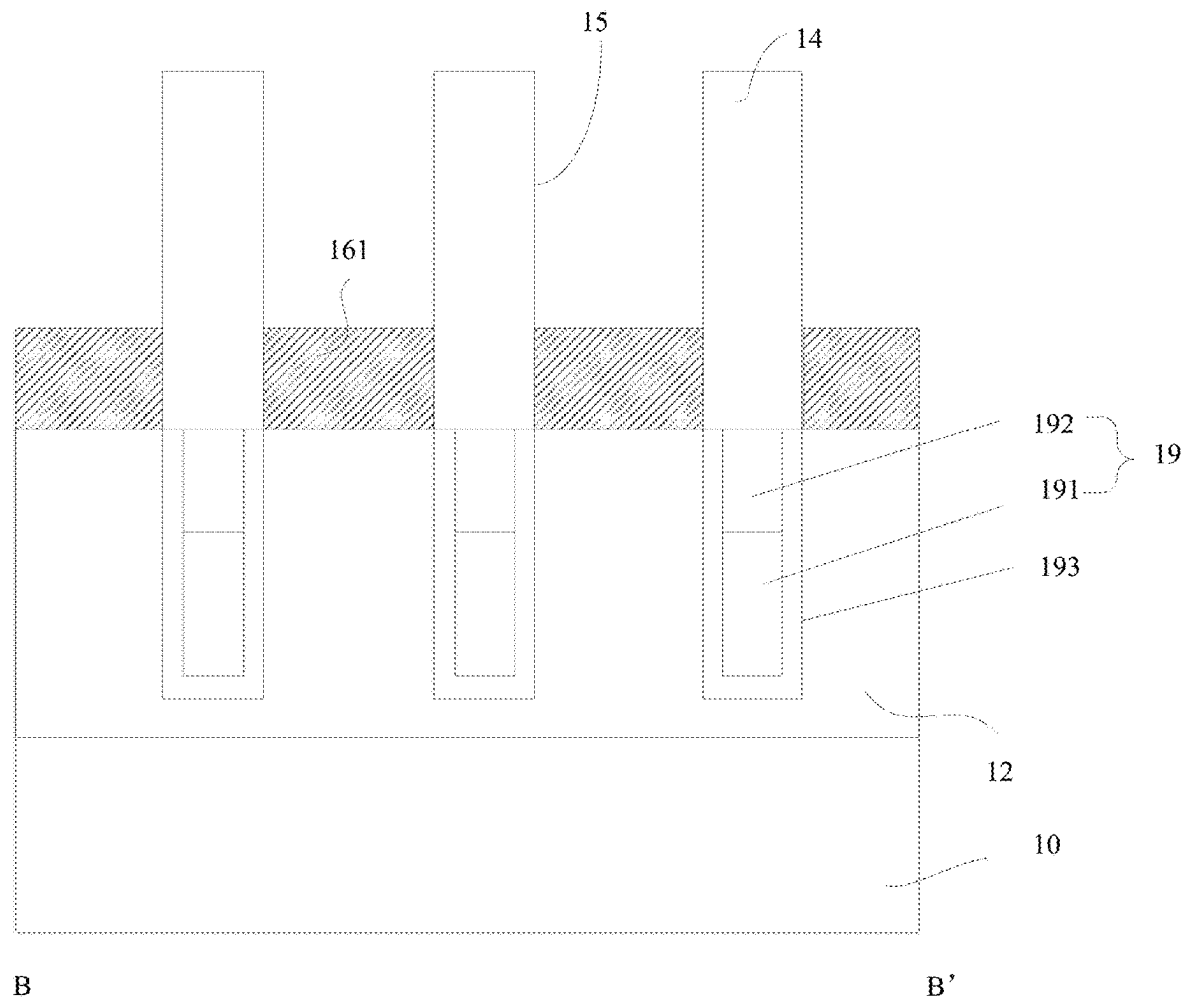


Figure 9B

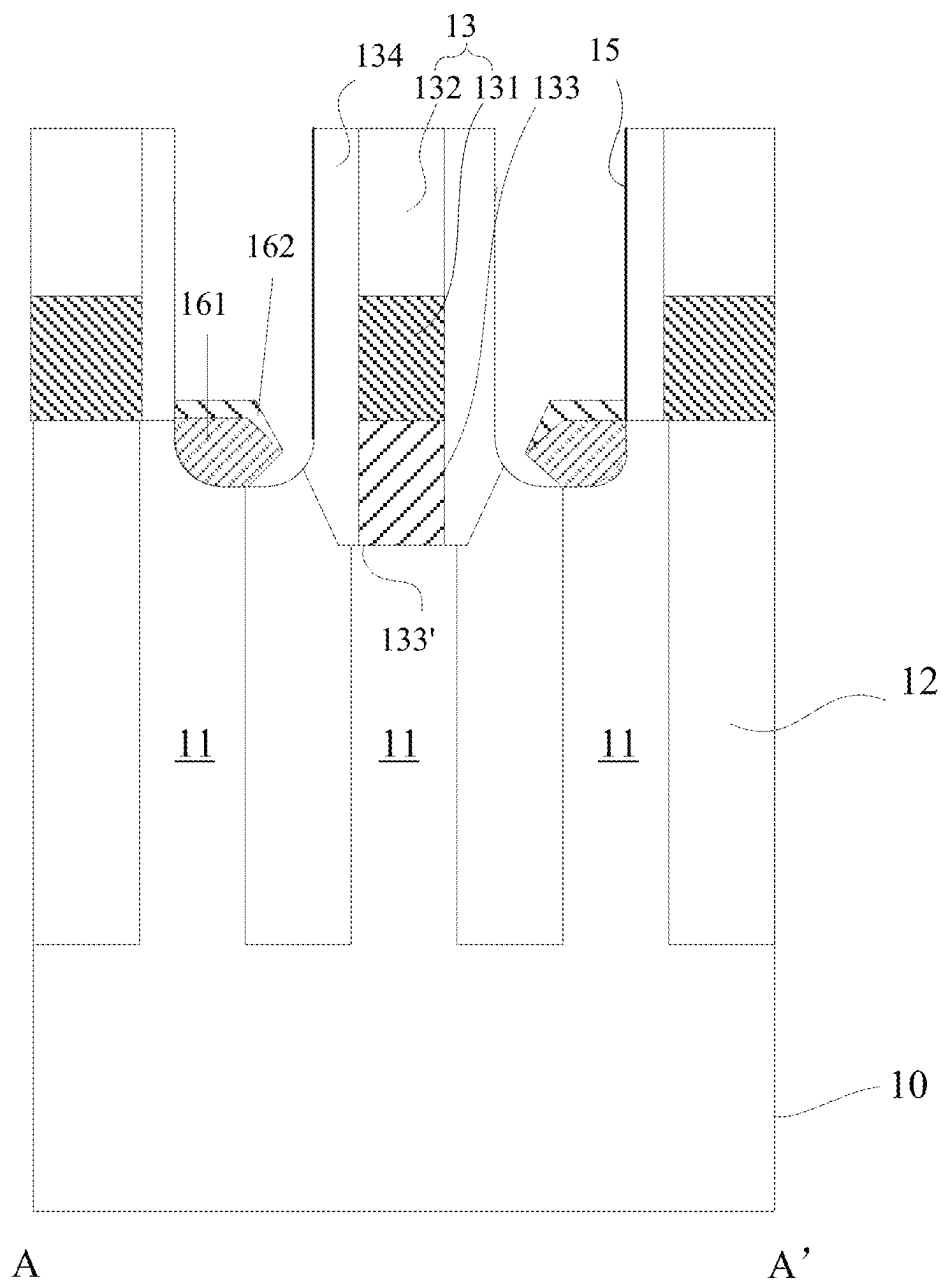


Figure 10A

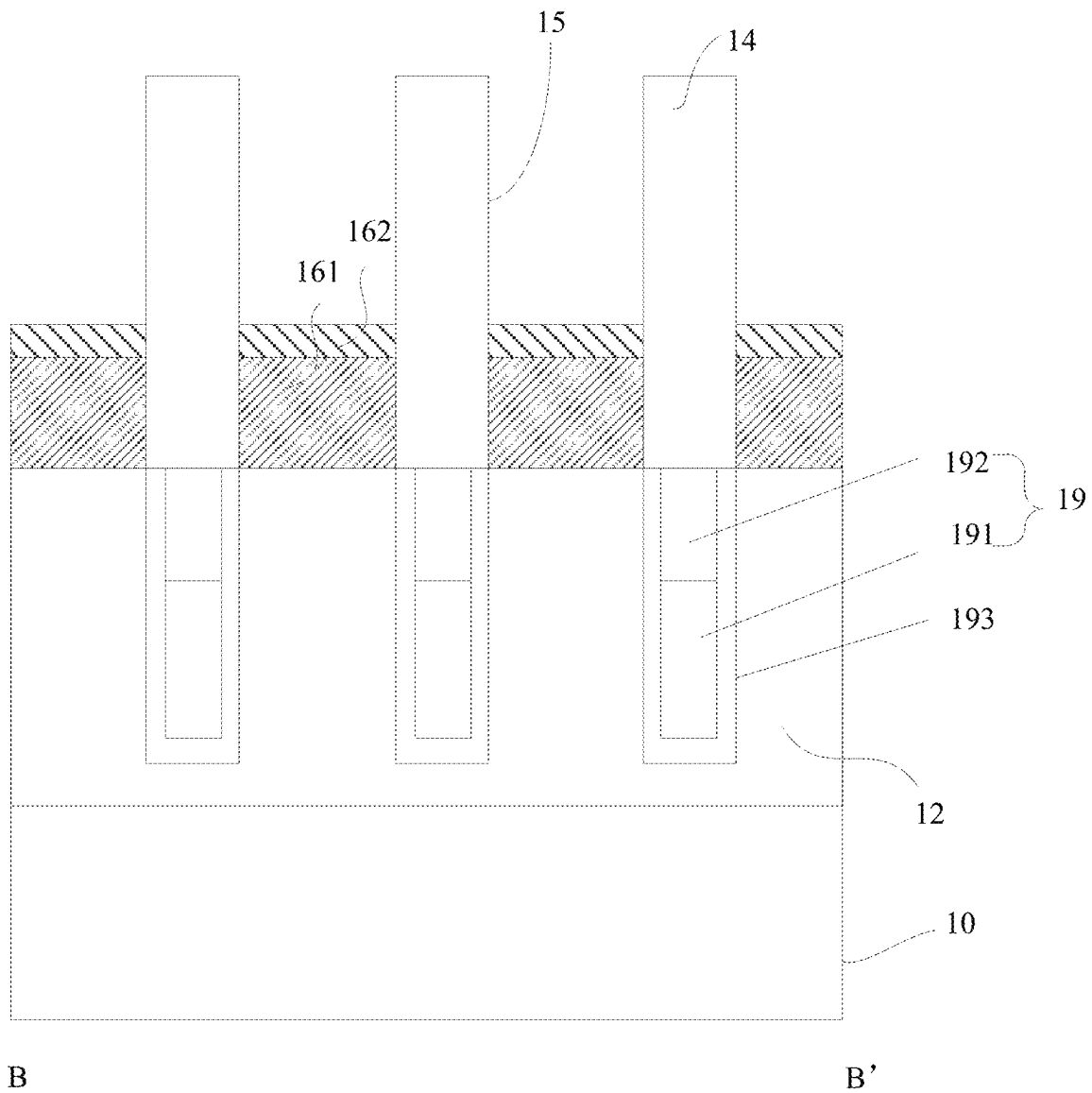


Figure 10B

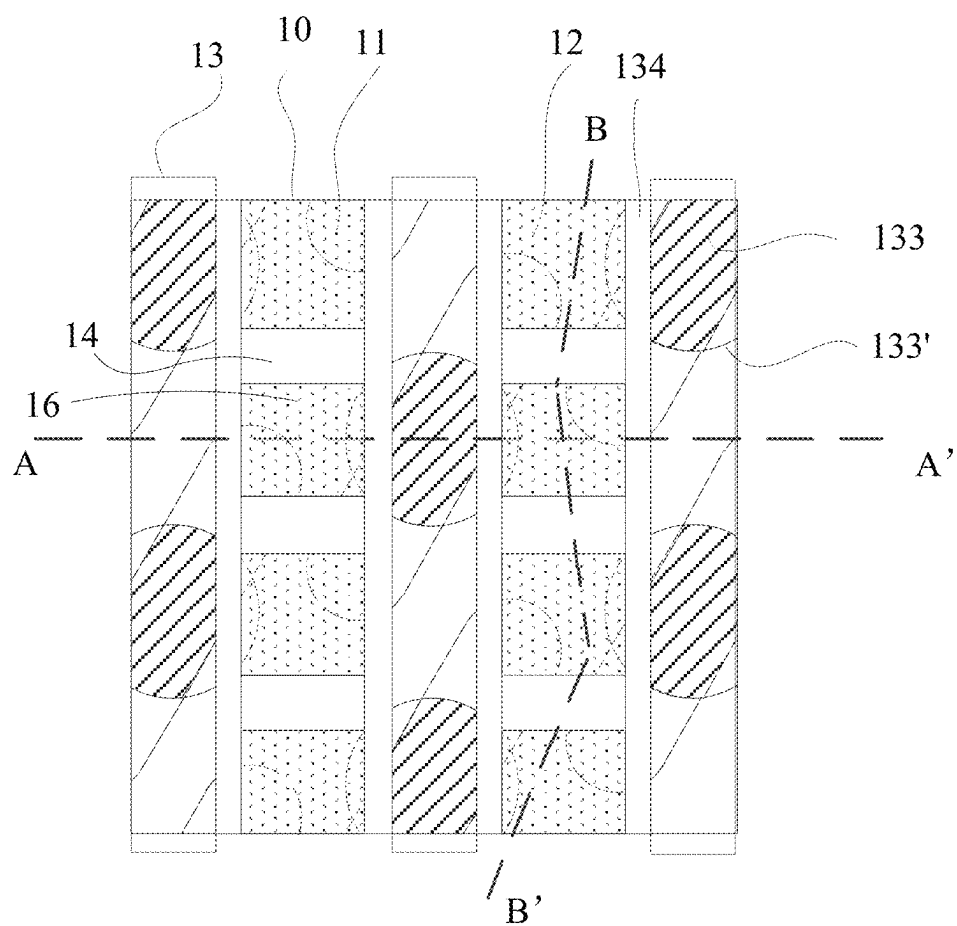


Figure 11A

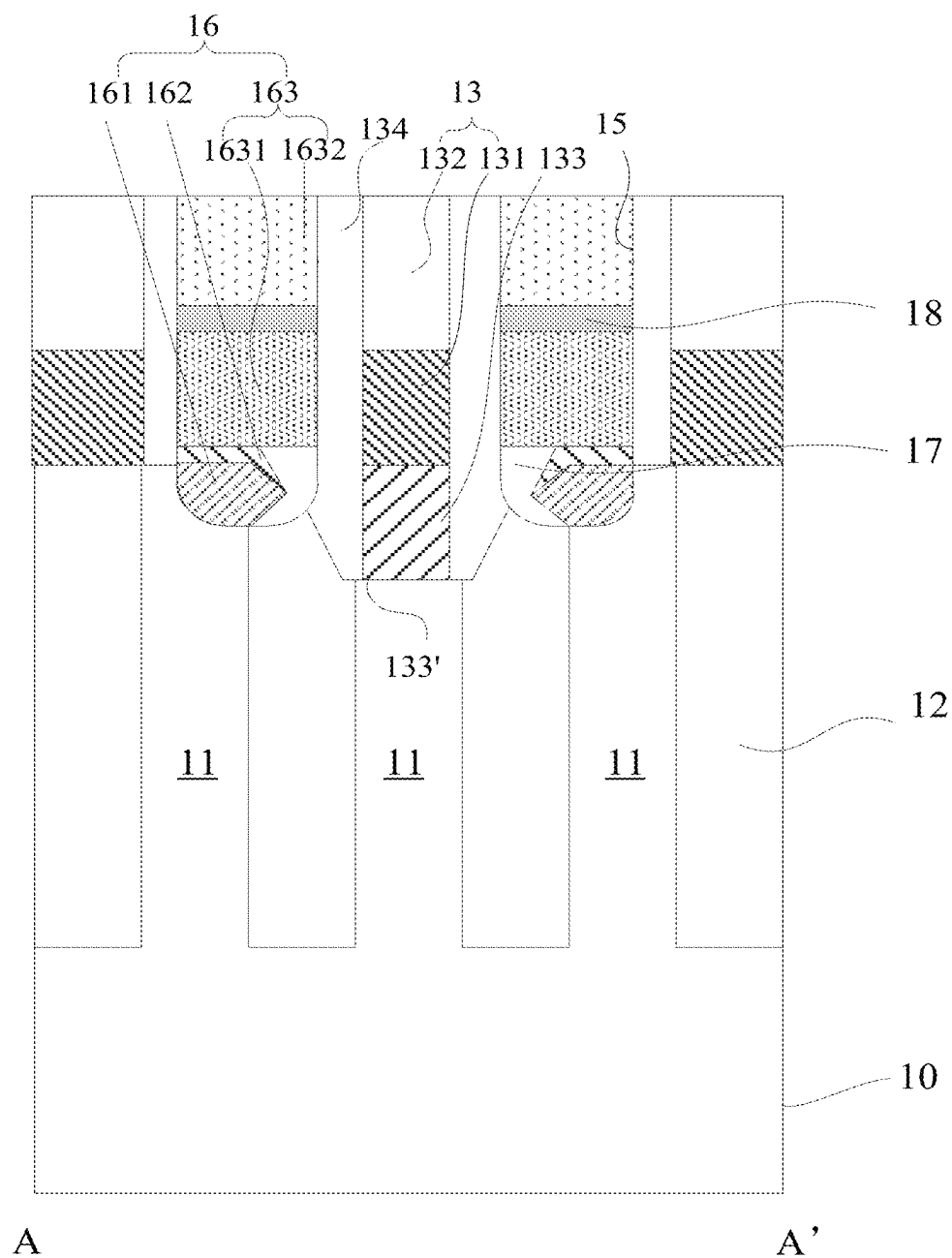


Figure 11B

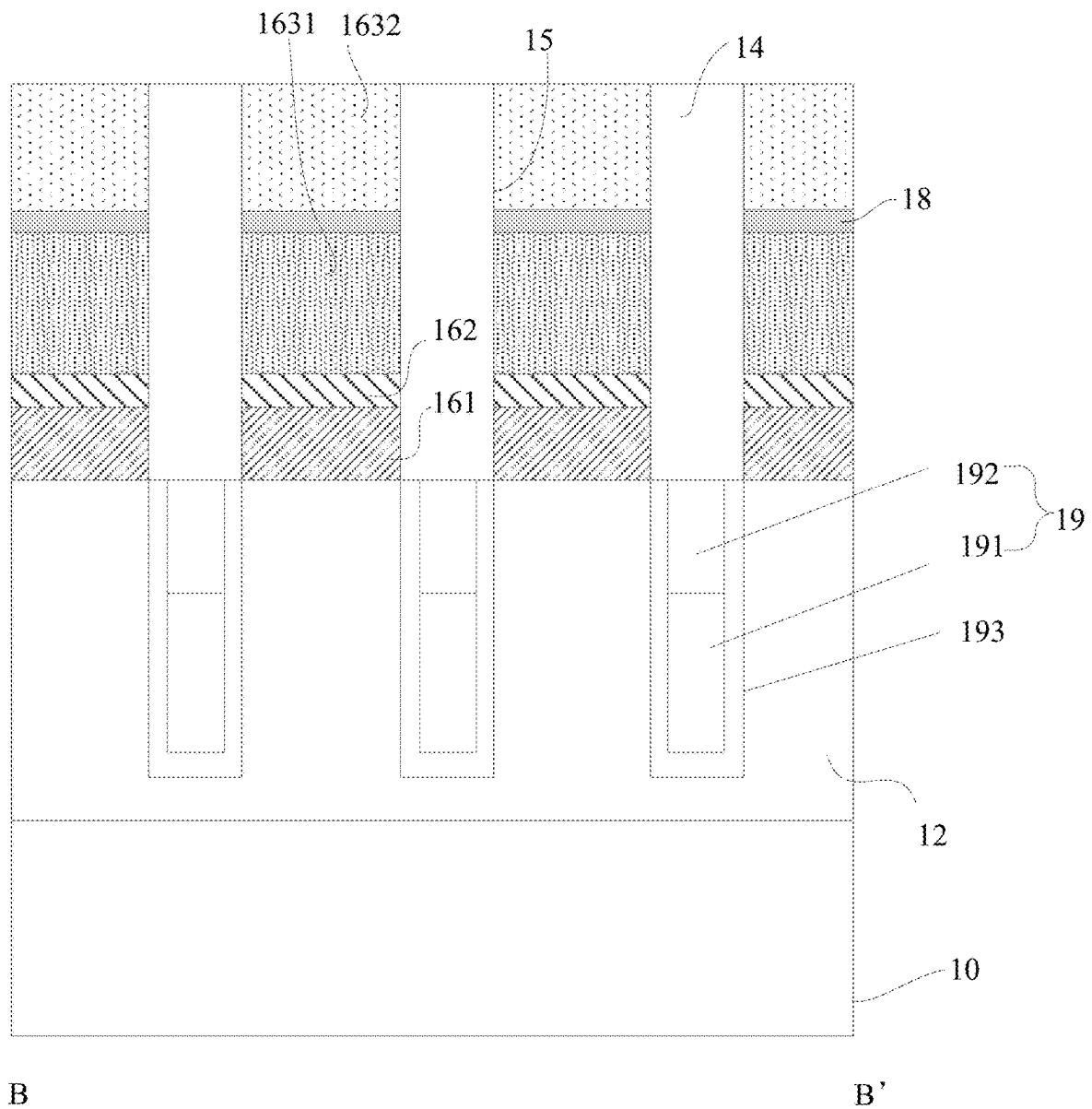


Figure 11C

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METHOD FOR FORMING SEMICONDUCTOR DEVICE HAVING AN AIR GAP BETWEEN A CONTACT PAD AND A SIDEWALL OF CONTACT HOLE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a divisional application of U.S. patent application Ser. No. 17/521,849, which claims the priority of Chinese patent application No. 202022770473.3, filed on Nov. 25, 2020 and entitled "Semiconductor device", and the priority of Chinese patent application No. 202011338461.1, filed on Nov. 25, 2020 and entitled "Semiconductor device and method for preparing semiconductor device", which are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

The present disclosure relates to the field of semiconductors, and in particular to a semiconductor device and a method for preparing a semiconductor device.

BACKGROUND OF THE INVENTION

With the development of technology, the level of integration of a semiconductor device gradually increases in order to satisfy a requirement for miniaturization of the semiconductor device. The increase of the level of integration requires a reduction in the dimension of a contact hole in the semiconductor device. A smaller contact hole will have a larger aspect ratio, and it will be difficult to control a morphology of a bottom part of the contact hole during a preparation process, causing deformation and oxidization of the contact hole. As a result, when a contact pad structure is formed in a contact hole having a larger aspect ratio, a bottom part of the contact pad structure and an active region exposed by the contact hole have poor contact, and the conductivity is relatively poor, thereby greatly affecting performance of a semiconductor device.

SUMMARY OF THE INVENTION

The technical problem to be solved by the present invention is how to realize effective contact between a contact pad structure and an active region in a semiconductor device with a contact hole having a larger aspect ratio to greatly improve a conductivity characteristic of the semiconductor device.

In order to solve the above-mentioned technical problem, the present disclosure provides a semiconductor device and a method for preparing a semiconductor device.

The first aspect of the present disclosure provides a semiconductor device, which may include:

- a substrate including at least an active region and a shallow trench isolation region;
- bit line structures located on the substrate and extending in a first direction;
- isolation fences located between adjacent bit line structures and arranged at intervals in the first direction;
- contact holes located in an area defined by adjacent bit line structures in a second direction and adjacent isolation fences in the first direction, where the second direction is perpendicular to the first direction, a bottom part of each of the contact holes extends into the

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- substrate and at least exposes part of the active region and part of the shallow trench isolation region; and
- a contact pad structure located in a respective contact hole, and including a first contact pad, a second contact pad adaptively covering the first contact pad, and a contact plug located on the second contact pad, where an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole.

The second aspect of the present disclosure provides a method for preparing a semiconductor device, which may include:

- forming, on a substrate, bit line structures extending in a first direction, where the substrate includes at least an active region and a shallow trench isolation region, the bit line structures are arranged at intervals in a second direction, and the second direction is perpendicular to the first direction;
- etching the substrate between adjacent bit line structures to form a contact opening that extends into the substrate and at least expose part of the active region and part of the shallow trench isolation region;
- forming isolation fences in the contact opening, wherein the isolation fences are arranged at intervals in the first direction to isolate the contact opening into a plurality of contact holes;
- performing, by using a germane gas, pretreatment on a bottom part of each of the contact holes;
- forming a first contact pad in a respective contact hole that has been subjected to the pretreatment;
- forming, on the first contact pad, a second contact pad adaptively covering the first contact pad, where an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole; and
- forming, in the contact holes, contact plugs, respectively.

BRIEF DESCRIPTION OF THE DRAWINGS

The scope of the present disclosure will be better understood from reading the following detailed description of exemplary embodiments in conjunction with the accompanying drawings. In the accompanying drawings:

FIG. 1 is a schematic top structural view of a semiconductor device provided in an embodiment of the present disclosure;

FIG. 2A is a schematic cross-sectional structural view along a line A-A' in FIG. 1 as provided in an embodiment of the present disclosure;

FIG. 2B is a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the present disclosure;

FIG. 2C is a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the present disclosure;

FIG. 3 is a schematic cross-sectional structural view along a line B-B' in FIG. 1 as provided in an embodiment of the present disclosure;

FIG. 4 is a schematic flowchart of a method for preparing a semiconductor device as provided in an embodiment of the present disclosure;

FIG. 5A is a schematic top structural view of a substrate 10;

FIG. 5B is a schematic cross-sectional structural view along a line A-A' in FIG. 5A;

FIG. 5C is a schematic cross-sectional structural view along the line B-B' in FIG. 5A;

FIG. 6A is a schematic top structural view of a semiconductor device provided with bit line structures;

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FIG. 6B is a schematic cross-sectional structural view along a line A-A' in FIG. 6A;

FIG. 6C is a schematic cross-sectional structural view along the line B-B' in FIG. 6B;

FIG. 7A is a schematic top structural view of a semiconductor device provided with a contact opening;

FIG. 7B is a schematic cross-sectional structural view along a line A-A' in FIG. 7A;

FIG. 7C is a schematic cross-sectional structural view along a line B-B' in FIG. 7A;

FIG. 8A is a schematic top structural view of a semiconductor device provided with contact holes 15;

FIG. 8B is a schematic cross-sectional structural view along a line A-A' in FIG. 8A;

FIG. 8C is a schematic cross-sectional structural view along a line B-B' in FIG. 8A;

FIG. 9A is a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with first contact pads 161;

FIG. 9B is a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with first contact pads 161;

FIG. 10A is a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with second contact pads 162;

FIG. 10B is a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with second contact pads 162;

FIG. 11A is a schematic top structural view of a semiconductor device provided with contact plugs 163;

FIG. 11B is a schematic cross-sectional structural view along a line A-A' in FIG. 11A; and

FIG. 11C is a schematic cross-sectional structural view along a line B-B' in FIG. 11A.

DETAILED DESCRIPTION OF THE EMBODIMENTS

In order to make the objects, technical solutions and advantages of the present disclosure clearer, the implementations of the present disclosure will be described in details below in conjunction with the accompanying drawings and embodiments, so as to fully understand and implement the implementation process of how to apply technical means to solve technical problems and achieve technical effects.

With the development of technology, the level of integration of a semiconductor device gradually increases in order to satisfy a requirement for miniaturization of the semiconductor device. The increase of the level of integration requires a reduction in the dimension of a contact hole in the semiconductor device. A smaller contact hole will have a larger aspect ratio, and it will be difficult to control a morphology of a bottom part of the contact hole during a preparation process, causing deformation and oxidization of the contact hole. As a result, when a contact pad structure is formed in a contact hole having a larger aspect ratio, a bottom part of the contact pad structure and an active region exposed by the contact hole have poor contact, and the conductivity is relatively poor, thereby greatly affecting performance of a semiconductor device.

Thus, the present disclosure provides a semiconductor device and a method for preparing a semiconductor device. The semiconductor device is provided with contact pad structures in contact holes. Each of the contact pad structures is configured to include a first contact pad, a second contact pad adaptively covering the first contact pad, and a contact plug located on the second contact pad. The first contact pad

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is in full contact with an active region in a substrate, which can effectively reduce contact resistance. Moreover, by providing contact pads in the form of a stacking structure, each of the first contact pads can function as a buffer layer for a respective second contact pad so as to avoid doped ions in the second contact pad from penetrating into the substrate and affecting conductivity. In addition, an air gap in each contact hole is formed between the first contact pad and a side wall on a side of the contact hole, which can effectively reduce an influence of the air gap on the conductivity, thereby facilitating significant improvement of a conductivity characteristic of the semiconductor device.

Embodiment One

With reference to FIG. 1, there is shown a schematic top structural view of a semiconductor device provided in an embodiment of the present disclosure. The semiconductor device includes a substrate 10, bit line structures 13, isolation fences 14, contact holes and contact pad structures 16.

The substrate 10 includes at least an active region 11 and a shallow trench isolation region 12.

The bit line structures 13 are located on the substrate 10 and extend in a first direction.

The isolation fences 14 are located between adjacent bit line structures 13 and arranged at intervals in the first direction.

The contact holes 15 are located in an area defined by adjacent bit line structures 13 in a second direction and adjacent isolation fences 14 in the first direction. The second direction is perpendicular to the first direction. A bottom part of each of the contact holes 15 extends into the substrate 10 and at least exposes part of the active region 11 and part of the shallow trench isolation region 12.

The contact pad structures 16 are located in the respective contact holes 15. Each of the contact pad structures 16 includes a first contact pad 161, a second contact pad 162 adaptively covering the first contact pad 161, and a contact plug 163 located on the second contact pad 162. An air gap 17 is formed between the first contact pad 161 and a side wall on a side of the respective contact hole 15.

The substrate 10 may include a semiconductor substrate. As a specific example, the substrate 10 may include a silicon substrate, a silicon-germanium substrate or a silicon substrate on an insulator. The substrate 10 may include the active region 11 and the shallow trench isolation region 12, and the shallow trench isolation region 12 may define the active region 11 into a plurality of areas.

In the embodiment of the present disclosure, the first direction may be represented by D1, and the second direction perpendicular to the first direction may be represented by D2. The bit line structures 13 extend in the direction D1, and the plurality of bit line structures are arranged at intervals in the direction D2.

As an example, each of the bit line structures 13 may include a bit line 131 and a hard mask layer 132 located on the bit line 131. Some of the bit line structures 13 may be in contact with the substrate 10 by means of bit line contact plugs 133. The bit line contact plugs 133 are arranged in respective bit line contact holes 133' inside the substrate 10, and each of the bit line contact holes 133' may have a transverse size greater than that of each of the bit line contact plugs 133. In addition, bit line spacers 134 may also be provided, each of which may adaptively cover a respective bit line structure 13 and fill in an air gap between a respective bit line contact plug 133 and a respective bit line contact hole 133'. An upper surface of the bit line contact

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plug 133 has the same height as an upper surface of the substrate 10. Referring particularly to FIG. 2A or 2B, FIG. 2A shows a schematic cross-sectional structural view along a line A-A' in FIG. 1 as provided in an embodiment of the present disclosure, and FIG. 2B shows a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the present disclosure. It should be noted that the size of the air gap 17 is enlarged in FIGS. 2A and 2B for convenience of view, and the size of the air gap 17 is not specifically limited.

As an example, the isolation fences 14 may be formed by depositing an insulation material. Specifically, the isolation fences 14 may be formed by depositing silicon nitride.

It should be noted that the isolation fences 14 may be formed before the contact holes 15 are formed, and may also be formed after the contact holes are formed. In the embodiment of the present disclosure, description is provided as the isolation fences 14 are formed after the contact holes 15 are formed. Referring particularly to FIG. 3, there is shown a schematic cross-sectional structural view along a line B-B' in FIG. 1 as provided in an embodiment of the present disclosure.

Referring back to FIG. 2A or 2B, the contact pad structures 16 are located in the respective contact holes 15. As an example, pretreatment may be performed on each of the contact holes 15 by using a germane gas before a respective first contact pad 161 is deposited, to remove impurities on an upper surface of the active region 11, so that the first contact pad 161 and the active region 11 are in good contact, thereby improving conductivity. The first contact pad 161 is then formed, by means of chemical vapor deposition, physical vapor deposition or the like, in a respective contact hole 15 that has been subjected to the pretreatment. It is possible to control, by means of selecting a suitable deposition condition, the first contact pad 161 to be in contact with part of a substrate surface of the respective contact hole 15. As one example, the first contact pad 161 may be in contact with part of a surface of each of the active region 11 and the shallow trench isolation region 12 in the respective contact hole 15 (not shown in the figures). As another example, the first contact pad 161 may be formed, by a selective epitaxial growth process, on the active region 11 in the respective contact hole 15 that has been subjected to the pretreatment. The first contact pad 161 is only in contact with the active region 11.

In addition, in the embodiment of the present disclosure, a polygonal first contact pad 161 having a planer top surface may be formed by controlling growth rates in different directions. In another embodiment, it is also possible to form a first contact pad 161 having an air gap between the same and a side wall on a side of a respective contact hole 15 and having an approximately elliptic cross-section. The first contact pad 161 may include a silicon germanium layer, and may also be an n-doped silicon germanium layer.

The second contact pad 162 adaptively covers the first contact pad 161. As an example, the second contact pad 162 may be formed on an upper surface of the first contact pad 161 by a selective epitaxial growth process, and an upper surface of the second contact pad 162 may be higher than a lower surface of the bit line structure 13.

The second contact pad 162 and the first contact pad 161 have different chemical compositions. For example, the second contact pad 162 may be formed from a material different from that of the first contact pad 161, and the second contact pad 162 may also be formed from the same material as that of the first contact pad 161 but with a

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different ion doping concentration. As a specific example, the second contact pad 162 may include a silicon phosphide layer.

The first contact pad 161 is formed by using the selective epitaxial growth process, and a bottom part of the first contact pad 161 is only in contact with the active region 11. After the second contact pad 162 is formed by using the selective epitaxial growth process, an air gap is formed between each of the first contact pad 161 and the second contact pad 162 and a side wall on a side of a respective contact hole 15. In such a structure, the first contact pad 161 may function as a buffer layer for the second contact pad 162, avoiding doped ions in the second contact pad 162 from penetrating into the substrate and affecting conductivity. Moreover, the air gap 17 in the contact hole 15 is located between the first contact pad 161 and a side wall on a side of the contact hole 15, which can effectively reduce an influence of the air gap 17 on the conductivity, thereby greatly improving performance of a semiconductor device.

In the embodiment of the present disclosure, since the first contact pad 161 is in contact with part of the substrate surface of the contact hole 15, an air gap will be formed between the first contact pad 161 and each of a side wall on a side of the respective contact hole 15 and a bottom surface of the respective contact hole 15 after the first contact pad 161 and the second contact pad 162 are both formed, and such an air gap may not be able to be completely filled when the contact plug 163 is subsequently formed by using a depositing process. As an example, the contact plug 163 may be formed on the second contact pad 162 by means of a chemical vapor deposition process or a physical vapor deposition process. Process conditions are controlled in such a way that the contact plug 163 only covers part of the surface of the respective second contact pad 162, and the air gap 17 is formed between each of the first contact pad 161 and the second contact pad 162 and the respective contact hole 15, as depicted in FIG. 2A for details.

As another example, the contact plug 163 may be formed on the respective second contact pad 162 by choosing a depositing process with a better step coverage characteristic, such that the contact plug 163 covers the second contact pad 162 and is in contact with part of the respective first contact pad 161, the air gap between each of the first contact pad 161 and the second contact pad 162 and the respective contact hole 15 is partially filled, and the air gap 17 is still reserved between the first contact pad 161 and a bottom part of the contact hole 15, as depicted in FIG. 2B for details.

As an example, the contact plug 163 may comprise a first contact insertion plug 1631 and a second contact insertion plug 1632 located on the first contact insertion plug 1631. An upper surface of the first contact insertion plug 1631 may be configured to be higher than an upper surface of a bit line 131, so that a contact interface between the first contact insertion plug 1631 and the second contact insertion plug 1632 is higher than the bit line 131, thereby avoiding ions at the contact interface from penetrating into the bit line 131 and affecting device performance.

In the embodiment of the present disclosure, the first contact insertion plug 1631 may include a polysilicon layer, a metal silicide or a metal layer, and the second contact insertion plug 1632 may include a metal layer. As a specific example, the first contact insertion plug 1631 may be a silicon phosphide layer, and the second contact insertion plug 1632 may be made of tungsten.

As another example, with reference to FIG. 2C, there is shown a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the

present disclosure. The contact pad structure **16** may also include a heavily-doped semiconductor layer **18** located between the first contact insertion plug **1631** and the second contact insertion plug **1632**, and contact resistance between the first contact insertion plug **1631** and the second contact insertion plug **1632** can be effectively reduced by providing the heavily-doped semiconductor layer **18**.

The above embodiment provides a semiconductor device with contact pad structures **16** in respective contact holes **15**. Each of the contact pad structures **16** is configured to include a first contact pad **161**, a second contact pad **162** adaptively covering the first contact pad **161**, and a contact plug **163** located on the second contact pad **162**. The first contact pad **161** is in full contact with an active region **11** in a substrate **10**, which can effectively reduce contact resistance. Moreover, by providing such contact pads in a stacking structure, the first contact pad **161** can function as a buffer layer for the second contact pad **162** so as to avoid doped ions in the second contact pad **162** from penetrating into the substrate **10** and affecting conductivity. In addition, an air gap **17** in each contact hole **15** is formed between the first contact pad **161**, the second contact pad **162**, the bit line spacer **134**, the shallow trench isolation region **12** and a side wall on a side of the contact hole **15**, which can effectively reduce an influence of the air gap **17** on the conductivity, thereby facilitating significant improvement of a conductivity characteristic of the semiconductor device.

Another aspect of the present disclosure further provides a method for preparing a semiconductor device. For details, reference can be made to the description in the following Embodiment Two.

Embodiment Two

With reference to FIG. 4, there is shown a schematic flowchart of the method for preparing a semiconductor device as provided in an embodiment of the present disclosure, and the method includes the following steps **S401** to **S407**.

At step **S401**, bit line structures **13** extending in a first direction are formed on a substrate **10**, where the substrate **10** includes at least an active region **11** and a shallow trench isolation region **12**, the bit line structures **13** are arranged at intervals in a second direction, and the second direction is perpendicular to the first direction.

At step **S402**, the substrate **10** between adjacent bit line structures **13** is etched to form a contact opening that extends into the substrate **10** and at least expose part of the active region **11** and part of the shallow trench isolation region **12**.

At step **S403**, isolation fences **14** are formed in the contact opening, where the isolation fences **14** are arranged at intervals in the first direction to isolate the contact opening into a plurality of contact holes **15**.

At step **S404**, pretreatment is performed on a bottom part of each of the contact holes **15** by using a germane gas.

At step **S405**, a first contact pad **161** is formed in a respective contact hole **15** that has been subjected to the pretreatment.

At step **S406**, a second contact pad **162** adaptively covering the first contact pad **161** is formed on the first contact pad **161**, where an air gap **17** is formed between the first contact pad **161** and a side wall on a side of the respective contact hole **15**.

At step **S407**, contact plugs **163** are formed in the contact holes **15**, respectively.

With reference to FIGS. 5A and 5B, FIG. 5A shows a schematic top structural view of a substrate **10**, and FIG. 5B

shows a schematic cross-sectional structural view along a line A-A' in FIG. 5A. The substrate **10** may include a semiconductor substrate. As an example, the substrate **10** may include a silicon substrate, a silicon-germanium substrate or a silicon substrate on an insulator. The substrate **10** may internally include the active region **11** and the shallow trench isolation region **12**, and the shallow trench isolation region **12** may define the active region **11** into a plurality of areas.

In the embodiment of the present disclosure, the first direction may be represented by D1, and the second direction perpendicular to the first direction may be represented by D2. In addition, with reference to FIG. 5C, there is shown a schematic cross-sectional structural view along a line B-B' in FIG. 5A. The substrate **10** may also be internally provided with buried word line structures **19**. The buried word line structures **19** extend in the direction D2. Each of the buried word line structures **19** may include a word line **191** and an insulation isolation layer **192** located on the word line **191**, and the buried word line structures **19** may be realized by means of a conventional approach in the art, which will not be described herein for brevity. In addition, dielectric layers **193** may also be arranged outside the buried word line structures **19**, respectively, and the dielectric layers **193** may be oxide layers or nitride layers.

Step **S401** may involve the following processes. Specifically, a hard mask layer is deposited and patterned on the substrate **10**. The patterned hard mask layer is used as a mask to etch the substrate **10**, so as to form bit line contact holes **133'** in the substrate **10**. A conductive material covering the bit line contact holes **133'** is deposited, where an upper surface of the conductive material may be the same height as an upper surface of the substrate **10**. A conductive layer covering the substrate **10** and the conductive material as well as a hard mask layer covering the conductive layer are deposited subsequently. And then the hard mask layer is patterned and the patterned hard mask layer is used as a mask to etch the conductive layer and the conductive material in the bit line contact holes **133'**, so as to form bit line structures **13** extending in the direction D1. The patterned conductive layer serves as the bit line **131**, and the patterned conductive material forms the bit line contact plug **133**. In addition, it is also possible to adaptively deposit bit line spacers **134**. With reference to FIGS. 6A-6C, FIG. 6A shows a schematic top structural view of bit line structures, FIG. 6B shows a schematic cross-sectional structural view along a line A-A' in FIG. 6A, and FIG. 6C shows a schematic cross-sectional structural view along a line B-B' in FIG. 6B. The bit line structure **13** may have the same line width as the bit line contact plug **133** in the direction D2. Each adaptively deposited bit line spacers **134** may also fill in an air gap between the bit line contact plugs **133** and the bit line contact holes **133'**.

Step **S402** may specifically involve: etching the substrate **10** between adjacent bit line structures **13** by means of a wet etching process or a dry etching process to form a contact opening that extends into the substrate **10** and at least expose part of the active region **11** and part of the shallow trench isolation region **12**. Referring particularly to FIGS. 7A-7C, FIG. 7A shows a schematic top structural view of a semiconductor device provided with a contact opening, FIG. 7B shows a schematic cross-sectional structural view along a line A-A' in FIG. 7A, and FIG. 7C shows a schematic cross-sectional structural view along a line B-B' in FIG. 7A. A line width of the contact opening in the direction D1 may be a distance between the bit line spacers **134** of adjacent bit line structures **13**. In another embodiment, the line width of

the contact opening in the direction D1 may also be smaller than the distance between the bit line spacers 134 of adjacent bit line structures 13.

Step S403 may involve the following processes. Specifically, isolation material layers 14' are deposited in the contact opening, where the isolation material layers 14' may be at the same height as the bit line spacers 134 on top surfaces of the respective bit line structures 13. A mask layer covering the bit line spacers 134 and the isolation material layers 14' is deposited and patterned. The patterned mask layer is then used as a mask to etch the isolation material layers 14', so as to form the isolation fences 14 that are arranged at intervals in the first direction. Adjacent bit line structures 13 in the direction D2 and adjacent isolation fences 14 in the direction D1 may isolate the contact opening into a plurality of contact holes 15. Referring particularly to FIGS. 8A-8C, FIG. 8A shows a semiconductor device provided with contact holes 15, FIG. 8B shows a schematic cross-sectional structural view along a line A-A' in FIG. 8A, and FIG. 8C shows a schematic cross-sectional structural view along a line B-B' in FIG. 8A.

The isolation material layers 14' may be insulation material layers. As a specific example, the isolation material layers 14' may be silicon nitride layers.

Step S404 may specifically involve: performing pretreatment on a bottom part of each of the contact holes 15 by using a germane gas. The active region 11 exposed in the contact holes 15 is extremely easily oxidized to form an insulation layer. For example, when the substrate 10 is a silicon layer, silicon oxide would be formed on an upper surface of the active region 11, and germanium ions in germane can damage a silicon-oxygen bond, to form volatile germanium oxide, such that the insulation material of the active region 11 can be cleaned, thereby facilitating great conductivity between each of the contact pads and the active region 11.

Step S405 may specifically involve: forming, by means of a chemical vapor deposition process or a physical vapor deposition process, the first contact pad 161 in a respective contact hole 15 that has been subjected to the pretreatment. It is possible to control, by means of selecting a suitable deposition condition, the first contact pad 161 to be in contact with part of a substrate surface of the respective contact hole 15. As one example, the first contact pad 161 may be in contact with part of a surface of each of the active region 11 and the shallow trench isolation region 12 in the respective contact hole 15 (not shown in the figures). As another example, the first contact pad 161 may be formed, by a selective epitaxial growth process, on the active region 11 in the respective contact hole 15 that has been subjected to the pretreatment. The first contact pad 161 is only in contact with the active region 11. As a specific example, the first contact pad 161 may be formed, by using the selective epitaxial growth process with silicane and germane as a precursor, on the active region 11 in the respective contact hole 15 that has been subjected to pretreatment. By using the selective epitaxial growth process, a bottom part of the first contact pad 161 is only in contact with the active region 11. Referring particularly to FIGS. 9A and 9B, FIG. 9A shows a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with first contact pads 161, and FIG. 9B shows a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with first contact pads 161.

Step S406 may specifically involve: forming, on the first contact pad 161 and by using the selective epitaxial growth process, the second contact pad 162 adaptively covering the

first contact pad 161. The second contact pad 162 and the first contact pad 161 have different chemical compositions. For example, the second contact pad 162 may be formed from a material different from that of the first contact pad 161, and the second contact pad 162 may also be formed from the same material as that of the first contact pad 161 but with a different ion doping concentration. As a specific example, the second contact pad 162 adaptively covering the first contact pad 161 is formed on the respective first contact pad 161 by using the selective epitaxial growth process with silicane and germane as a precursor. Referring particularly to FIG. 10A, there is shown a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with second contact pads 162. As an example, an upper surface of each of the second contact pads 162 may be higher than a lower surface of each of the bit lines 131. FIG. 10B also shows a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with second contact pads 162. After each second contact pad 162 is formed by using the selective epitaxial growth process, an air gap may be formed between each of the first contact pad 161 and the second contact pad 162 and a side wall on a side of the respective contact hole 15.

It should be noted that since the contact hole 15 has a larger aspect ratio and the first contact pad 161 is in contact with part of the substrate surface of the respective contact hole 15, an air gap will be formed between the first contact pad 161 and each of a side wall on a side of the respective contact hole 15 and a bottom surface of the respective contact hole 15 after the first contact pad 161 and the second contact pad 162 are both formed, but such an air gap may not be able to be completely filled when the contact plugs 163 are subsequently formed by using a depositing process.

As an example, step S407 may specifically involve: forming a respective contact plug 163 on a respective second contact pad 162 by means of a chemical vapor deposition process or a physical vapor deposition process. Process conditions are controlled in such a way that the contact plug 163 only covers part of the surface of the respective second contact pad 162, and the air gap 17 is formed between each of the first contact pads 161 and the second contact pads 162 and the respective contact hole 15, as depicted in FIG. 11B for details. As another example, step S407 may specifically involve: forming a respective contact plug 163 on a respective second contact pad 162 by choosing a depositing process with a better step coverage characteristic, such that the contact plug 163 covers the second contact pad 162, and is in contact with part of the respective first contact pad 161, the air gap between each of the first contact pad 161 and the second contact pad 162 and the respective contact hole 15 is partially filled, and the air gap 17 is still reserved between the first contact pad 161 and a bottom part of the respective contact hole 15, as depicted in FIG. 2B in Embodiment One for details. Hereinafter, descriptions will be made based on the example that a contact plug 163 covering part of a surface of a respective second contact pad 162 is formed in a respective contact hole 15. With reference to FIGS. 11A-11C, FIG. 11A shows a schematic top structural view of a semiconductor device provided with contact plugs 163, and FIG. 11B shows a schematic cross-sectional structural view along a line A-A' in FIG. 11A. An upper surface of a first contact insertion plug 1631 may be higher than an upper surface of a bit line 131, so that a contact interface between the first contact insertion plug 1631 and a second contact insertion plug 1632 is higher than the bit line 131, thereby avoiding ions at the contact interface from penetrating into

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the bit line **131** and affecting device performance. FIG. **11C** shows a schematic cross-sectional structural view along a line B-B' in FIG. **11A**.

Step **S407** may also specifically involve: depositing, by means of the chemical vapor deposition process or the physical vapor deposition process, a polysilicon layer, a metal silicide or a metal layer in a respective contact hole **15** as its first contact insertion plug **1631**; and depositing a metal layer in the respective contact hole **15** to form its second contact insertion plug **1632** covering the first contact insertion plug **1631**. The first contact pad **161**, the second contact pad **162**, the first contact insertion plug **1631** and the second contact insertion plug **1632** may form the contact pad structure **16**.

As an example, the first contact insertion plug **1631** may be a silicon phosphide layer, and the second contact insertion plug **1632** may be made of tungsten. When the first contact insertion plug **1631** is made of the same material as the second contact pad **162**, the growth rate of the first contact insertion plug **1631** may be greater than that of the second contact pad **162**.

It should be noted that after the polysilicon layer, the metal silicide or the metal layer is formed in the respective contact hole **15** as the first contact insertion plug **1631**, a heavily-doped semiconductor layer **18** covering the first contact insertion plug **1631** may also be deposited.

The above embodiment provides a method for preparing a semiconductor device. In this method, pretreatment is performed on a bottom part of each contact hole **15** by using a germane gas, first contact pads **161** and second contact pads **162** adaptively covering the respective first contact pads **161** grow on an active region **11** by using a selective epitaxial growth process, to form contact plugs **163** in the contact holes **15**, respectively, such that the first contact pads **161** are in full contact with the active region **11** in a substrate **10**, which can effectively reduce contact resistance. Moreover, by providing contact pads in the form of a stacking structure, each of the first contact pads can function as a buffer layer for a respective second contact pad **162** so as to avoid doped ions in the second contact pad **162** from penetrating into the substrate **10** and affecting conductivity. In addition, an air gap **17** in each contact hole **15** is formed between the first contact pad **161** and a side wall on a side of the contact hole **15**, which can effectively reduce an influence of the air gap on the conductivity, thereby facilitating significant improvement of a conductivity characteristic of the semiconductor device.

Although various implementations are disclosed herein, it should be understood that they have been presented for facilitating the understanding of the present disclosure, rather than limiting the scope of the present disclosure. A skilled person in the art can make any modification and change to the form and details of the implementation without departing from the spirit and scope of the present disclosure, and the scope of the present disclosure shall be defined in the appended claims.

What is claimed is:

1. A method for preparing a semiconductor device, comprising:

forming bit line structures extending in a first direction on a substrate, wherein the substrate comprises at least an active region and a shallow trench isolation region, the

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bit line structures are arranged at intervals in a second direction, and the second direction is perpendicular to the first direction;

etching the substrate between adjacent bit line structures to form a contact opening that extends into the substrate and at least expose part of the active region and part of the shallow trench isolation region;

forming isolation fences in the contact opening, wherein the isolation fences are arranged at intervals in the first direction to isolate the contact opening into a plurality of contact holes;

performing pretreatment on a bottom part of each of the contact holes;

forming a plurality of first contact pads in the respective contact holes;

forming a second contact pad adaptively covering the first contact pad, wherein an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole; and

forming contact plugs in the contact holes, respectively.

2. The method according to claim 1, wherein the first contact pad is in contact with part of a substrate surface.

3. The method according to claim 2, further comprises: forming the first contact pad by using a selective epitaxial growth process.

4. The method according to claim 1, further comprises: forming the second contact pad adaptively covering the first contact pad, wherein the air gap is among of first contact pad, the second contact pad and the side wall on a side of the respective contact hole.

5. The method according to claim 1, further comprises: forming a contact plug covering the respective second contact pad and in contact with part of a surface of the respective first contact pad.

6. The method according to claim 1, further comprises: forming a polysilicon layer, a metal silicide or a metal layer in the respective contact hole as first contact insertion plug, and depositing a metal layer to form second contact insertion plug covering the first contact insertion plug.

7. The method according to claim 6, wherein the first contact insertion plug is made of the same material as the second contact pad, and a growth rate of the first contact insertion plug is greater than a growth rate of the second contact pad.

8. The method according to claim 6, wherein after the step of forming the polysilicon layer, the metal silicide or the metal layer in the respective contact hole as the first contact insertion plug, the method further comprises:

depositing a heavily-doped semiconductor layer covering the first contact insertion plug.

9. The method according to claim 6, wherein an upper surface of the second contact pad is higher than a lower surface of the bit line, and an upper surface of the first contact insertion plug is higher than an upper surface of the bit line.

10. The method according to claim 1, further comprises: depositing a conductive layer covering the substrate and a hard mask layer covering the conductive layer; and patterning the conductive layer and the hard mask layer, wherein the patterned conductive layer serves as a bit line, and the patterned hard mask layer is located on the bit line.

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